



**Swami Keshvanand Institute of Technology,
Management & Gramothan, Jaipur**
(An Autonomous Institute Affiliated to Rajasthan
Technical University, Kota)

I-Mid Term Examination, March-2025

Prog./Semester:	B. Tech.-VI	Branch:	IT
Course Title:	DIP	Course Code:	6IT3-01
Duration:	1.5 Hours	Maximum Marks:	20
Session : First		Roll No.:	
Instructions if any:		NA	

PART -A (short answer type questions)

(All questions are compulsory)

(All questions are compulsory)																												
Q. No.	Question					Max. Marks	CO (s)	Bloom's Level																				
1	How to digital Image represent also Calculate the storage required for a 1024x1024 binary image.					2	1	3																				
2	Explain Fourier transform? Discuss its properties.					2	2	2																				
3	Compute the median value of marked pixels of image using 3x3 mask.					2	2	3																				
<table><tr><td>18</td><td>22</td><td>33</td><td>25</td><td>32</td><td>24</td></tr><tr><td>34</td><td><u>128</u></td><td><u>24</u></td><td><u>172</u></td><td><u>26</u></td><td>23</td></tr><tr><td>22</td><td>19</td><td>32</td><td>31</td><td>28</td><td>26</td></tr></table>									18	22	33	25	32	24	34	<u>128</u>	<u>24</u>	<u>172</u>	<u>26</u>	23	22	19	32	31	28	26		
18	22	33	25	32	24																							
34	<u>128</u>	<u>24</u>	<u>172</u>	<u>26</u>	23																							
22	19	32	31	28	26																							

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO (s)	Bloom's Level
4	Categorize the step-by-step process involved in digital image processing.	4	1	4
5	Discuss Sampling and Quantization in digital images.	4	1	2

PART- C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any one Question)

Q. No.	Question	Max. Marks	CO (s)	Bloom's Level								
6	Generate the histogram equalized image for the input Image as below:	6	2	5								
	Gray level				0	1	2	3	4	5	6	7
	No. of Pixels				8	10	10	2	12	16	4	2
7	Demonstrate the different types of intensity transform functions used in Digital Image Processing.	6	2	5								



Solution of Question Paper
I Mid-Term Examination, March -2025

Prog./Sem.: B. Tech.-VI	Course Title: DIP	Course Code: 6IT3-01
Duration: 1.5 hours	Date: 12.03.2025 Session : FIRST	Max Marks: 20
Submitted By: Dr. S.R. Dogiwal		

S. No.	Course Outcome	Percentage Coverage
CO1	Identify the fundamental elements of an Image and Describe the need of Digital Image Processing	50%
CO2	Understand different types of image transformation techniques and their properties.	50%
CO3	Use various noise models and Calculate the values for restoration and degradation.	0%
CO4	Analyze and Evaluate various image compression techniques.	0%
CO5	Integrate and Demonstrate various image transformation and segmentation techniques.	0%

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	-
2	BL2- Understand	30%
3	BL3- Apply	20%
4	BL4- Analyze	20%
5	BL5- Evaluate	30%
6	BL6- Create	-

-- Question Paper Solution --



Solution of Question Paper
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Prog./Sem.: B. Tech.-VI	Course Title: DIP	Course Code: 6IT3-01
Duration: 1.5 hours	Date: 12.03.2025 Session : FIRST	Max Marks: 20
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Q.1 How to digital Image represent also Calculate the storage required for a 1024x1024 binary image.

Solution: A digital image is represented using a grid of pixels, where each pixel corresponds to a specific location in the image and contains information about color and brightness. The representation of a digital image depends on factors such as color depth and file format.

Now, let's calculate the storage required for a 1024x1024 binary image. In a binary image, each pixel is represented using 1 bit. Therefore, the total number of bits required can be calculated as:

Total bits = width × height × bits per pixel

Total bits= width × height × bits per pixel

Given that the image is 1024x1024 pixels and each pixel is represented using 1 bit, we have:

Total bits=1024×1024×1

Total bits=1,048,576 bits

To convert bits to bytes, we divide by 8 (since 1 byte = 8 bits):

Total bytes = 1,048,576 bits/8

Total bytes =131,072 bytes

To convert bytes to kilobytes (KB), we divide by 1024:

Total KB=131,072/1024

Total KB=128 KB

Therefore, a 1024x1024 binary image requires approximately 128 kilobytes (KB) of storage.

Q. 2 Explain Fourier transform? Discuss its properties.

Solution: Fourier Transform is a mathematical model which helps to transform the signals between two different domains, such as transforming signal from frequency domain to spatial domain or vice versa.

Properties are : Separable, Symmetric, Translation, Periodicity, Rotation, Distribution, Scaling

Q. 3 Compute the median value of marked pixels of image using 3x3 mask.

18	22	33	25	32	24
34	<u>128</u>	<u>24</u>	<u>172</u>	<u>26</u>	23
22	19	32	31	28	26



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Solution:

Original Pixels Values

18	22	33	25	32	24
34	128	24	172	26	23
22	19	32	31	28	26

Median Value of Marked Pixels

18	22	33	25	32	24
34	24	31	31	26	23
22	19	32	31	28	26

So Original Pixel 128, 24, 172, 26 are replaced
With median Values 24, 31, 31, 26 respectively.

Q.4 Examine the step-by-step process involved in digital image processing

Solution: The step-by-step process involved in digital image processing can vary depending on the specific task or application. However, a general overview of the process typically includes the following steps:

A digital image is an image composed of picture elements, also known as pixels, each with finite, discrete quantities of numeric representation for its intensity or gray level that is an output from its two-dimensional functions fed as input by its spatial coordinates denoted with x, y on the x-axis and y-axis, respectively. Step 1: Image Acquisition: The image is captured by a sensor (eg. Camera), and digitized if the output of the camera or sensor is not already in digital form, using analogue-to-digital convertor

Step 2: Image Enhancement: The process of manipulating an image so that the result is more suitable than the original for specific applications. The idea behind enhancement techniques is to bring out details that are



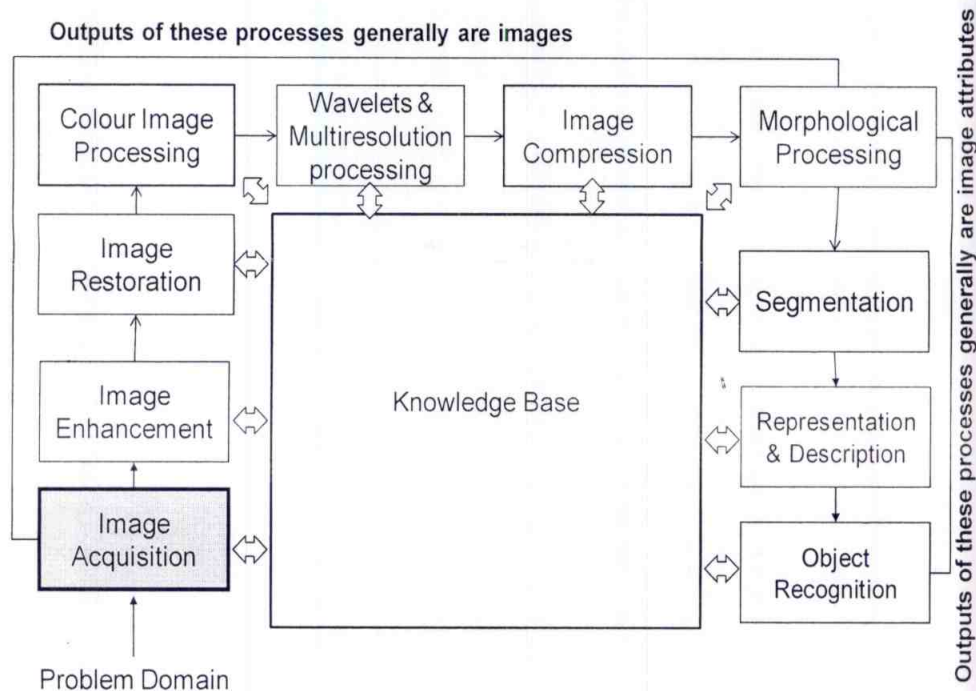
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hidden, or simple to highlight certain features of interest in an image.

Step 3: Image Restoration: Improving the appearance of an image and Tend to be mathematical or probabilistic models. Enhancement, on the other hand, is based on human subjective preferences regarding what constitutes a "good" enhancement result.

Fundamental Steps in Digital Image Processing:



Step 4: Colour Image Processing Use the colour of the image to extract features of interest in an image

Step 5: Wavelets: are the foundations of representing images in various degrees of resolution. It is used for image data compression.

Step 6: Compression: Techniques for reducing the storage required to save an image or the bandwidth required to transmit it. Techniques for reducing the storage required to save an image or the bandwidth required to transmit it.

Step 7: Image Morphological Process: Segmentation procedures partition an image into its constituent parts or objects. The more accurate the segmentation, the more likely recognition is to succeed.

Step 8: Image Segmentation: Segmentation procedures partition an image into its constituent parts or objects. Important Tip: The more accurate the segmentation, the more likely recognition is to succeed.

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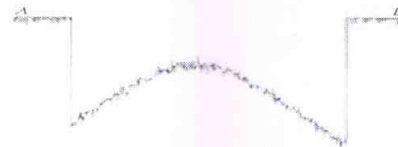
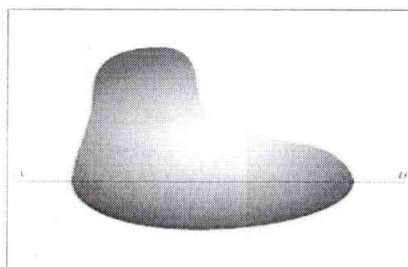
Step 9: Representation and Description: Representation: Make a decision whether the data should be represented as a boundary or as a complete region. It is almost always follows the output of a segmentation stage. Boundary Representation: Focus on external shape characteristics, such as corners and inflections, Region Representation: Focus on internal properties, such as texture or skeleton shape

Step 10: Recognition and Interpretation: Recognition: the process that assigns label to an object based on the information provided by its description.

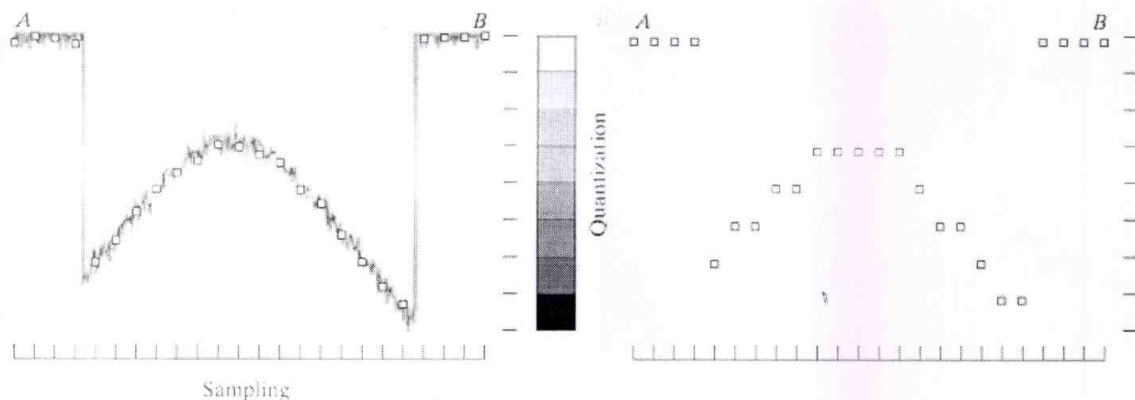
Step 11: Knowledge Base Knowledge about a problem domain is coded into an image processing system in the form of a knowledge database.

Q. 5 Discuss Sampling and Quantization in digital images.

Solution: : In order to become suitable for digital processing, an image function $f(x,y)$ must be digitized both spatially and in amplitude. Typically, a frame grabber or digitizer is used to sample and quantize the analogue video signal. Hence in order to create an image which is digital, we need to covert continuous data into digital form. There are two steps in which it is done: Sampling and Quantization



The sampling rate determines the spatial resolution of the digitized image, while the quantization level determines the number of grey levels in the digitized image. A magnitude of the sampled image is expressed as a digital value in image processing. The transition between continuous values of the image function and its digital equivalent is called quantization





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- To convert it to digital form, we have to sample the function in both coordinates and in amplitude. Digitizing the coordinate values is called sampling. Digitizing the amplitude values is called quantization.
- In practice, the method of sampling is determined by the sensor arrangement used to generate the image.
- When an image is generated by a single sensing element combined with mechanical motion, sampling is accomplished by selecting the number of individual mechanical increments at which we activate the sensor to collect data.
- Hence highest sampling rate is achieved with single sensing element.

Q. 6 Generate the histogram equalized image for the input Image as below:

Gray level	0	1	2	3	4	5	6	7
No. of Pixels	8	10	10	2	12	16	4	2

Solution:

Soln: The equalization process for given image

r_k	P_k	Cumulative values	$\frac{\text{Cumulative}}{\text{Total}} \times (L-1)$	Round off to nearest grey level
0	8	8	$8/64 \times 7 = 0.875$	1
1	10	18	$18/64 \times 7 = 1.968$	2
2	10	28	$28/64 \times 7 = 3.0625$	3
3	2	30	$30/64 \times 7 = 3.2812$	3
4	12	42	$42/64 \times 7 = 4.5937$	5
5	16	58	$58/64 \times 7 = 6.3437$	6
6	4	62	$62/64 \times 7 = 6.78125$	7
7	2	64	$64/64 \times 7 = 7$	7

Show the final distribution of grey scale pixels where the pixels 28 and 30 corresponding to the level 2 and 3 are mapped to level 3 in resultant image as similar - 62264 - mapped to 7



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Q.7 Demonstrate the different types of intensity transform function used in Digital Image Processing.

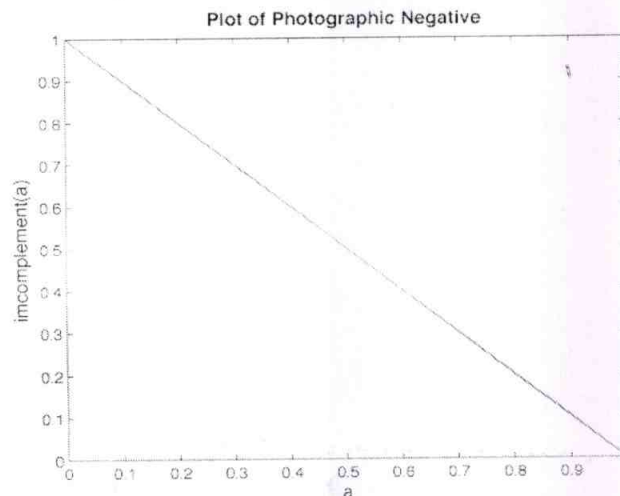
Solution: Intensity transformation functions in digital image processing are used to modify the pixel intensity values of an image. These functions can be categorized based on their purposes and effects on the image. Here are the different types of intensity transform functions:

Intensity Transformation When you are working with gray-scale images, sometimes you want to modify the intensity values. For instance, you may want to reverse black and the white intensities or you may want to make the darks darker and the lights lighter. An application of intensity transformations is to increase the contrast between certain intensity values so that you can pick out things in an image.

Photographic Negative

The Photographic Negative is probably the easiest of the intensity transformations to describe. Assume that we are working with grayscale double arrays where black is 0 and white is 1. The idea is that 0's become 1's, 1's become 0's, and any gradients in between are also reversed. In intensity, this means that the true black becomes true white and vice versa.

$f^*(x,y) = f(x,y) - (L-1)$ for gray scale image L is 256 $f^*(x,y)$ is transformed pixel.



Log Transformations:

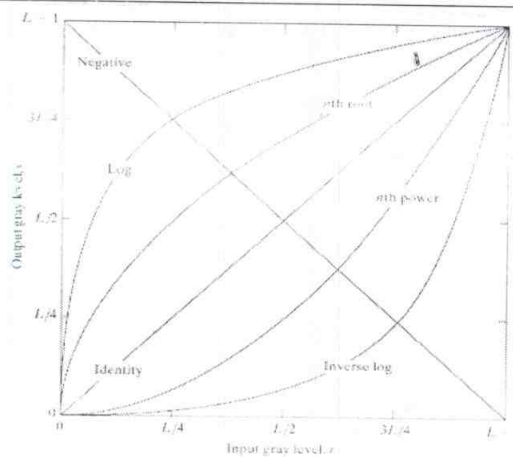
$$s = c \log(1+r), \quad c = \text{const}, \quad r \geq 0$$

Maps a narrow range of low intensity values in the input into a wider range of output levels. The opposite is true for higher values of input levels.

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FIGURE 3.3 Some basic gray-level transformation functions used for image enhancement.



Power-Law (Gamma) transformation

$$s = cr^\gamma, \quad c, \gamma - \text{positive constants}$$

curve the grayscale components either to brighten the intensity (when $\gamma < 1$)

or darken the intensity (when $\gamma > 1$).

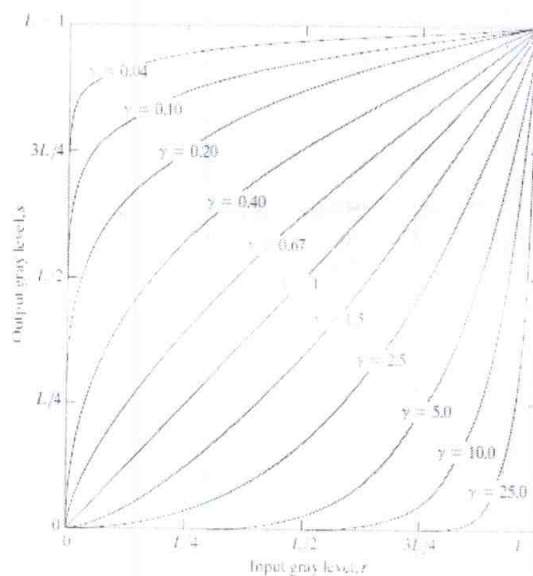


FIGURE 3.6 Plots of the equation $s = cr^\gamma$ for various values of γ ($c = 1$ in all cases).



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Duration:	1.5 Hours	Maximum Marks:	20
Session: First		Roll No.:	
Instructions if any:			

PART -A (short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	Write one difference and one similarity between classification and regression.	2	CO1	BL2
2	How do you compute Support and Confidence for finding the association rules? Give example	2	CO2	BL2
3	What is " support vector " in SVM? Explain with help of any example?	2	CO1	BL1

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max Marks	CO(s)	Bloom's Level												
4	Apply the K-Nearest Neighbour's algorithm to any classification problem and explain how you would determine the optimal value of k . Discuss the impact of choosing a very small or very large value of k on the model's performance.	4	CO1	BL3												
5	The table below shows the relationship between the number of hours studied (x) and the marks obtained (y) by a group of students: <table><tr><td>Hours Studied (X)</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>Marks Obtained (y)</td><td>35</td><td>50</td><td>65</td><td>75</td><td>90</td></tr></table> (a) Find the equation of the regression line. (b) Predict the marks obtained if a student studies for 7 hours. Interpret the slope of the regression line in the context of this problem.	Hours Studied (X)	2	4	6	8	10	Marks Obtained (y)	35	50	65	75	90	4	CO1	BL3
Hours Studied (X)	2	4	6	8	10											
Marks Obtained (y)	35	50	65	75	90											

PART- C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any one Question)

Q. No	Question	Max. Marks	CO(s)	Bloom's Level																																																												
6	The following distance matrix represents the pairwise distances between five data points (P, Q, R, S, T): <table><tr><td></td><td>P</td><td>Q</td><td>R</td><td>S</td><td>T</td></tr><tr><td>P</td><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td>Q</td><td>3</td><td>0</td><td></td><td></td><td></td></tr><tr><td>R</td><td>8</td><td>5</td><td>0</td><td></td><td></td></tr><tr><td>S</td><td>10</td><td>7</td><td>2</td><td>0</td><td></td></tr><tr><td>T</td><td>12</td><td>9</td><td>4</td><td>3</td><td>0</td></tr></table> (a) Perform hierarchical clustering using complete linkage. Show the steps of merging clusters and compute the distances at each step. (b) Draw the dendrogram for the clustering result.		P	Q	R	S	T	P	0					Q	3	0				R	8	5	0			S	10	7	2	0		T	12	9	4	3	0	6	CO2	BL4																								
	P	Q	R	S	T																																																											
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T	12	9	4	3	0																																																											
7	What are the valid association rules generated from the frequent itemsets using the Apriori algorithm. <table><tr><td>1</td><td>Milk</td><td>Egg</td><td>Bread</td><td>Butter</td></tr><tr><td>2</td><td>Milk</td><td>Butter</td><td>Egg</td><td>Ketchup</td></tr><tr><td>3</td><td>Bread</td><td>Butter</td><td>Ketchup</td><td></td></tr><tr><td>4</td><td>Milk</td><td>Bread</td><td>Butter</td><td></td></tr><tr><td>5</td><td>Bread</td><td>Butter</td><td>Cookies</td><td></td></tr><tr><td>6</td><td>Milk</td><td>Bread</td><td>Butter</td><td>Cookies</td></tr><tr><td>7</td><td>Milk</td><td>Cookies</td><td></td><td></td></tr><tr><td>8</td><td>Milk</td><td>Bread</td><td>Butter</td><td></td></tr><tr><td>9</td><td>Bread</td><td>Butter</td><td>Egg</td><td>Cookies</td></tr><tr><td>10</td><td>Milk</td><td>Butter</td><td>Bread</td><td></td></tr><tr><td>11</td><td>Milk</td><td>Bread</td><td>Butter</td><td></td></tr><tr><td>12</td><td>Milk</td><td>Bread</td><td>Cookies</td><td>Ketchup</td></tr></table> Consider the minimum support threshold (33%). And Generate association rules from the frequent itemsets and prune based on the minimum confidence threshold (60%).	1	Milk	Egg	Bread	Butter	2	Milk	Butter	Egg	Ketchup	3	Bread	Butter	Ketchup		4	Milk	Bread	Butter		5	Bread	Butter	Cookies		6	Milk	Bread	Butter	Cookies	7	Milk	Cookies			8	Milk	Bread	Butter		9	Bread	Butter	Egg	Cookies	10	Milk	Butter	Bread		11	Milk	Bread	Butter		12	Milk	Bread	Cookies	Ketchup	6	CO2	BL4
1	Milk	Egg	Bread	Butter																																																												
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Prog./Sem.: BTech / VI Semester	Course Title: Machine Learning	Course Code: 6IT4-02
Duration: 1.5 hours	Date: 11-03-25 Session: First	Max Marks: 20
Submitted By: Praveen Kumar Yadav, Ms. Nikhar Bhatnagar		

S. No.	Course Outcome	Percentage Coverage
CO1	Demonstrate a variety of supervised machine-learning models and show case their applications.	60%
CO2	Illustrate various applications of unsupervised machine learning models.	40%
CO3	Recognize diverse machine-learning algorithms through statistical learning techniques.	--
CO4	Discuss the methodologies of semi-supervised learning and reinforcement learning.	--
CO5	Employ the concepts of Neural Networks within the realm of Machine Learning.	--

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	10%
2	BL2- Understand	20%
3	BL3- Apply	40%
4	BL4- Analyze	30%
5	BL5- Evaluate	--
6	BL6- Create	--



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-- Question Paper Solution --

PART-A

Q.1 Answer: Difference and Similarity Between Classification and Regression

Difference:

- Classification predicts discrete labels (e.g., spam vs. not spam), while regression predicts continuous values (e.g., house prices).

Similarity:

- Both are supervised learning techniques that use labeled data to train models for making predictions.

Q.2 Answer: Computing Support and Confidence for Association Rules

Support:

- Proportion of transactions containing an itemset. For example, if 5 out of 10 transactions contain {bread, milk}, support is 50%.

Confidence:

- Measures the likelihood of an itemset given another. For example, if 4 out of 5 transactions containing {bread} also contain {milk}, confidence is 80%.

Q.3 Answer: Support Vector in SVM

Support Vector:

- Data points closest to the decision boundary (hyperplane) in SVM. They influence the boundary's position and orientation. For instance, in a binary classification of cats vs. dogs, support vectors are the images of cats and dogs nearest to the separating line.



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PART-B

Q.4 Answer: The K-Nearest Neighbors (KNN) algorithm is a simple and intuitive classification method used in machine learning. It classifies a data point based on the majority class among its k nearest neighbors in the feature space. Here's how you can apply the KNN algorithm to a classification problem and determine the optimal value of k :

Steps to Apply KNN Algorithm

1. Choose the Number of Neighbors (k):

- Start with an odd value for k to avoid ties in classification. Common starting points are $k=3$ or $k=5$.

2. Select a Distance Metric:

- Common distance metrics include Euclidean, Manhattan, and Minkowski distances. Euclidean distance is the most frequently used.

3. Normalize the Data:

- Ensure that all features are on a similar scale to prevent features with larger scales from dominating the distance calculations.

4. Train the Model:

- KNN is a lazy learning algorithm, meaning it doesn't explicitly learn a model. Instead, it stores the training data and makes predictions based on the stored data.

5. Make Predictions:

- For a new data point, calculate the distance to all training points.
- Identify the k nearest neighbors.
- Assign the class that is most frequent among these k neighbors to the new data point.

Determining the Optimal Value of k

The choice of k is crucial as it directly impacts the model's performance. Here's how to determine the optimal value:

1. Cross-Validation:

- Use techniques like k -fold cross-validation to evaluate the model's performance for different values of k .
- Split the data into k subsets (folds), train the model on $k-1$ folds, and test it on the remaining fold. Repeat this process k times, each time using a different fold as the test set.



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- Calculate the average accuracy or error rate across all folds for each value of k.

2. Elbow Method:

- Plot the error rate or accuracy against different values of k.
- Look for the “elbow point” where the error rate starts to increase or the accuracy starts to decrease significantly. This point often indicates the optimal value of k.

3. Heuristic Techniques:

- Start with a small value of k (e.g., k=3) and incrementally increase it while monitoring the model's performance.
- Choose the value of k that provides the best trade-off between bias and variance.

Impact of k on Model Performance

• **Small Values of k:**

- A very small value of k (e.g., k=1) can lead to a model that is highly sensitive to noise in the training data. The model may overfit the training data, capturing noise and outliers, which reduces its generalization capability.
- The decision boundary will be highly irregular and complex, fitting the training data too closely.

• **Large Values of k:**

- A very large value of k can lead to a model that is too simplistic and may underfit the data. The decision boundary will be smoother, but it may not capture the underlying patterns in the data.
- The model may ignore important features and fail to capture the complexity of the data, leading to higher bias and lower accuracy.

Q.5 Answer:

(a) Finding the Equation of the Regression Line

To find the equation of the regression line, we use the formula for simple linear regression:

$$y = mx + c$$

where:

- m is the slope of the line.
- c is the y-intercept.



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I Mid-Term Examination, MARCH-2025

Prog./Sem.: BTech / VI Semester	Course Title: Machine Learning	Course Code: 6IT4-02
Duration: 1.5 hours	Date: 11-03-25 Session: First	Max Marks: 20
Submitted By: Praveen Kumar Yadav, Ms. Nikhar Bhatnagar		

The slope m and intercept c can be calculated using the following formulas:

$$m = \frac{N(\sum xy) - (\sum x)(\sum y)}{N(\sum x^2) - (\sum x)^2}$$

$$c = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{N(\sum x^2) - (\sum x)^2}$$

where N is the number of data points.

First, we calculate the necessary sums:

X	Y	XY	X ²
2	35	70	4
4	50	200	16
6	65	390	36
8	75	600	64
10	90	900	100

Sum of X ($\sum x$) = 2 + 4 + 6 + 8 + 10 = 30

Sum of Y ($\sum y$) = 35 + 50 + 65 + 75 + 90 = 315

Sum of XY ($\sum xy$) = 70 + 200 + 390 + 600 + 900 = 2160

Sum of X² ($\sum x^2$) = 4 + 16 + 36 + 64 + 100 = 220



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Number of data points (N) = 5

Now, we substitute these values into the formulas for m and c:

$$m = \frac{5(2160) - (30)(315)}{5(220) - (30)^2} = \frac{10800 - 9450}{1100 - 900} = \frac{1350}{200} = 6.75$$

$$c = \frac{(315)(220) - (30)(2160)}{5(220) - (30)^2} = \frac{69300 - 64800}{1100 - 900} = \frac{4500}{200} = 22.5$$

So, the equation of the regression line is:

$$y = 6.75x + 22.5$$

(b) Predicting the Marks Obtained for 7 Hours of Study

To predict the marks obtained if a student studies for 7 hours, we substitute $x=7$ into the regression equation:

$$y = 6.75(7) + 22.5$$

$$y = 47.25 + 22.5$$

$$y = 69.75$$

$$y = 69.75$$

So, the predicted marks obtained for 7 hours of study are approximately 69.75.

Interpretation of the Slope

The slope of the regression line is 6.75. This means that for every additional hour of study, the predicted marks increase by 6.75 points. In other words, each extra hour of study is associated with an average increase of 6.75 marks in the exam score.

PART-C

Q.6 Answer: (a) **Hierarchical Clustering Using Complete Linkage**

Step 1: Initialization

Start with each data point as its own cluster:

- Clusters: {P}, {Q}, {R}, {S}, {T}



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Step 2: First Merge

Find the pair of clusters with the smallest complete linkage distance. The complete linkage distance between two clusters is the maximum distance between any data point in the first cluster and any data point in the second cluster.

From the distance matrix:

- The smallest distance is between R and S with a distance of 2.

Merge R and S:

- New Cluster: {R, S}
- Remaining Clusters: {P}, {Q}, {T}

Step 3: Update Distance Matrix

Calculate the complete linkage distances between the new cluster {R, S} and the remaining clusters:

- Distance between {R, S} and {P}: $\max(8, 10) = 10$
- Distance between {R, S} and {Q}: $\max(5, 7) = 7$
- Distance between {R, S} and {T}: $\max(4, 3) = 4$

Updated distance matrix:

	P	Q	{R, S}	T
P	0	3	10	12
Q	3	0	7	9
{R, S}	10	7	0	4
T	12	9	4	0

Step 4: Second Merge

Find the pair of clusters with the smallest complete linkage distance.

- The smallest distance is between P and Q with a distance of 3.

Merge P and Q:



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- New Cluster: {P, Q}
- Remaining Clusters: {R, S}, {T}

Step 5: Update Distance Matrix

Calculate the complete linkage distances between the new cluster {P, Q} and the remaining clusters:

- Distance between {P, Q} and {R, S}: $\max(10, 7) = 10$
- Distance between {P, Q} and {T}: $\max(12, 9) = 12$

Updated distance matrix:

	{P, Q}	{R, S}	T
{P, Q}	0	10	12
{R, S}	10	0	4
T	12	4	0

Step 6: Third Merge

Find the pair of clusters with the smallest complete linkage distance.

- The smallest distance is between {R, S} and T with a distance of 4.

Merge {R, S} and T:

- New Cluster: {R, S, T}
- Remaining Clusters: {P, Q}

Step 7: Update Distance Matrix

Calculate the complete linkage distances between the new cluster {R, S, T} and the remaining cluster {P, Q}:

- Distance between {P, Q} and {R, S, T}: $\max(10, 12) = 12$

Updated distance matrix:



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	{P, Q}	{R, S, T}
{P, Q}	0	12
{R, S, T}	12	0

Step 8: Final Merge

Merge the remaining clusters {P, Q} and {R, S, T}:

- Final Cluster: {P, Q, R, S, T}

(b) Dendrogram

The dendrogram represents the hierarchical clustering process. Here is the structure of the dendrogram based on the merges:



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Q.7 Answer:

Step 1: **Generate 1-itemsets**

First, we count the occurrences of each item in the dataset.

Item	Support Count	Support (%)
Milk	9	75%
Egg	3	25%
Bread	10	83%
Butter	10	83%
Ketchup	3	25%
Cookies	5	42%

Step 2: **Prune 1-itemsets**

We prune the 1-itemsets based on the minimum support threshold of 33%.

Item	Support Count	Support (%)
Milk	9	75%
Bread	10	83%
Butter	10	83%
Cookies	5	42%

Step 3: **Generate 2-itemsets-Next**, we generate 2-itemsets from the frequent 1-itemsets and count their occurrences.

2-itemset	Support Count	Support (%)
{Milk, Bread}	7	58%
{Milk, Butter}	7	58%
{Milk, Cookies}	3	25%
{Bread, Butter}	9	75%
{Bread, Cookies}	4	33%
{Butter, Cookies}	3	25%



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Step 4: Prune 2-itemsets

We prune the 2-itemsets based on the minimum support threshold of 33%.

2-itemset	Support Count	Support (%)
{Milk, Bread}	7	58%
{Milk, Butter}	7	58%
{Bread, Butter}	9	75%
{Bread, Cookies}	4	33%

Step 5: Generate 3-itemsets

Next, we generate 3-itemsets from the frequent 2-itemsets and count their occurrences.

3-itemset	Support Count	Support (%)
{Milk, Bread, Butter}	6	50%
{Milk, Bread, Cookies}	2	17%
{Milk, Butter, Cookies}	1	8%
{Bread, Butter, Cookies}	3	25%

Step 6: Prune 3-itemsets: We prune the 3-itemsets based on the minimum support threshold of 33%.

3-itemset	Support Count	Support (%)
{Milk, Bread, Butter}	6	50%

Step 7: Generate Association Rules

Finally, we generate association rules from the frequent itemsets, and prune based on the minimum confidence threshold of 60%.

Frequent 3-item set = $I \Rightarrow \{Milk, Bread, Butter\}$

Non-Empty Subset are - S

$S = \{\{Milk\}, \{Bread\}, \{Butter\}, \{Milk, Bread\}, \{Milk, Butter\}, \{Bread, Butter\}\}$

How to form Association rule -

- For every non-empty subset S of I, the association rule is -

$$S \rightarrow (I - S)$$

$$IF \text{ Support}(I) / \text{Support}(S) \geq \text{min_confidence}$$

- Min support = 33% and Min_confidence = 60%



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Rule 1: {Milk} → {Bread, Butter}

Confidence = Support(I) / Support(S) = Support({Milk, Bread, Butter}) / Support(Milk) = $6/12 / 9/12 = 6/9 = 66.67\%$

- Valid rule.

Rule 2: {Bread} → {Milk, Butter}

Confidence = $(6/12) / (10/12) = 6/10 = 60\% \geq 60\%$

- Valid rule.

Rule 3: {Butter} → {Milk, Bread}

Confidence = $(6/12) / (10/12) = 6/10 = 60\% \geq 60\%$ Valid rule.

Rule 4: {Milk, Bread} → {Butter}

Confidence = Support({Milk, Bread, Butter}) / Support({Milk, Bread}) = $6/7 \approx 85.7\% \geq 60\%$

- Valid rule.

Rule 5: {Milk, Butter} → {Bread}

Confidence = Support({Milk, Bread, Butter}) / Support({Milk, Butter}) = $6/7 \approx 85.7\% \geq 60\%$

- Valid rule

Rule 6: {Bread, Butter} → {Milk}

Confidence = Support({Milk, Bread, Butter}) / Support({Bread, Butter}) = $6/9 \approx 66.67\% \geq 60\%$

- Valid rule

Association Rule	Confidence (%)
{Milk, Bread} → {Butter}	85.70%
{Milk, Butter} → {Bread}	85.70%
{Bread, Butter} → {Milk}	66.67%
{Milk} → {Bread, Butter}	66.67%
{Bread} → {Milk, Butter}	60.00%
{Butter} → {Milk, Bread}	60.00%

Likewise generate all association rules from qualified pruned 2 itemset also.



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Course Title:	ISS	Course Code:	6IT4-03
Duration:	1.5 Hours	Maximum Marks:	20
Session : Second		Roll No.:	

PART -A (Short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	Define the terms: cryptography & cryptanalysis.	2	1	1
2	Analyse the applicability of CFB to both stream and block ciphers? Draw the block diagram of the same.	2	2	4
3	Explain the core principles of security, and how do they contribute to the overall protection of information systems?	2	1	4

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	Describe the design principles of block cipher in detail.	4	2	2
5	Apply Playfair cipher technique to encrypt the message "MY NAME IS SKIT". Use keyword "PLAYFAIR SECURITY".	4	1	3

PART- C

(Descriptive/Analytical/Problem solving/Design questions solving)

(Attempt any one Question)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
6	Describe DES algorithm with the help of suitable block diagrams. Compute the 4- bit output of 6-bit input (101101) using S-Box.	6	2	3
7	Discuss the concept of key distribution in symmetric key cryptography. Compute the shared secret key using the Diffie-Hellman Key Exchange Algorithm with the provided values: Given: Prime number $p=5$, Base $g=3$, Alice's private key $a=6$, Bob's private key $b=15$.	6	2	3



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S. No.	Course Outcome	Percentage Coverage
CO1	Understand Security Fundamentals and Encryption Techniques.	40
CO2	Apply modern block cipher with modes.	60
CO3	Apply Public Key Cryptosystems for Secure Communication.	-
CO4	Analyze the fundamentals concepts Cryptographic Hash Functions and MACs.	-
CO5	Illustrate proficiency in key management and web engineering security.	-

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	10
2	BL2- Understand	20
3	BL3- Apply	50
4	BL4- Analyze	20
5	BL5- Evaluate	-
6	BL6- Create	-

-- Question Paper Solution --

Q.1 Define the terms: cryptography & cryptanalysis.

Ans. Cryptography is the practice and study of techniques used to secure communication and information. It is primarily concerned with creating systems (algorithms) that ensure the confidentiality, integrity, authentication, and non-repudiation of data. Cryptography involves encoding or encrypting messages so that only authorized parties can read or understand them.

Cryptanalysis is the art of breaking cryptographic systems and algorithms. It involves analyzing encrypted data with the goal of finding weaknesses in the cryptographic techniques used and gaining access to the encrypted information without the key.

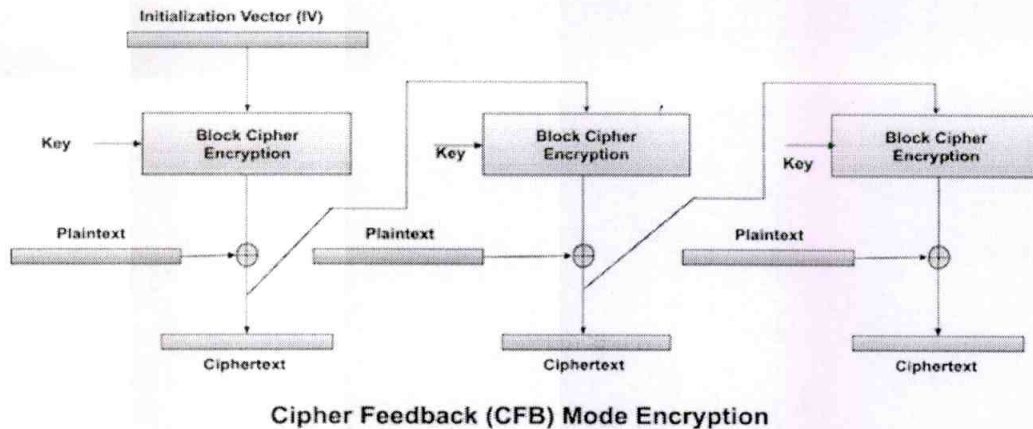
Q.2 Analyse the applicability of CFB to both Stream and Block Ciphers? Draw the block diagram of the same.

Ans. Stream Cipher-Like Behavior: Even though it uses a block cipher, CFB mode can encrypt data one bit or byte at a time, making it function like a stream cipher.

Block Cipher-Based Security: It leverages the strong cryptographic properties of a block cipher to create a secure key stream.

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Q.3 Explain the core principles of security, and how do they contribute to the overall protection of information systems?

Ans. Five security services to prevent security attacks—

- i. Data Confidentiality
- ii. Data Integrity
- iii. Authentication
- iv. Non-Repudiation
- v. Access Control

A process (or a device compilation such a process) that is designed to identify, avoid, or restore from a security attack. The mechanisms are divided into those that are performed in a definite protocol layer, including TCP or an application-layer protocol.

These principles ensure that sensitive data remains secure, systems are resilient to threats, and access is controlled and monitored.

Q.4 Describe the design principles of block cipher in detail.

Ans. Designing a block cipher involves several key principles to ensure the algorithm's security, efficiency, and usability. Here are the core design principles of block ciphers:

1. Confusion

- Goal: Make the relationship between the plaintext and ciphertext as complex as possible.
- Description: The ciphertext should not reveal any useful information about the plaintext. This is achieved by mixing up the bits of the plaintext in a way that is not easily reversible without the key. The use of substitution steps in block ciphers, such as the S-boxes (substitution boxes), helps achieve confusion.

2. Diffusion



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- Goal: Spread the influence of each bit of the plaintext over many bits of the ciphertext.
- Description: This makes patterns in the plaintext less discernible in the ciphertext. A small change in the plaintext should affect many parts of the ciphertext. Diffusion is often achieved by permutation operations such as p-boxes (permutation boxes) or bitwise shifting, which rearrange bits across the block.

3. Key Schedule

- Goal: Derive a series of round keys from the main secret key.
- Description: A secure block cipher requires that the key schedule (the process of generating round key from the original key) should prevent the attacker from inferring the secret key from the round keys. The schedule should also ensure that each round key is used in a manner that makes it difficult for attackers to guess or correlate keys.

4. Rounds

- Goal: Repeatedly apply confusion and diffusion processes to increase security.
- Description: Block ciphers typically operate by applying several rounds of encryption. Each round consists of substitution, permutation, and possibly XOR operations, ensuring that the encryption process is not easily reversible without the key. The number of rounds is critical—too few rounds make the cipher vulnerable to attacks, and too many rounds may result in inefficiency.

5. Inversion (Reversibility)

- Goal: The encryption process must be invertible to enable decryption.
- Description: A cipher is invertible if, given the ciphertext and the key, it is possible to recover the original plaintext. This is important for both encryption and decryption to work correctly and efficiently.

6. Security Against Known Attacks

- Goal: Design the cipher to withstand various cryptanalytic attacks.
- Description: The block cipher must resist common attacks, including:
 - Differential Cryptanalysis: Uses the differences between pairs of plaintexts to analyze the cipher.
 - Linear Cryptanalysis: Uses linear approximations to describe the relationship between plaintext, ciphertext, and the key.
 - Meet-in-the-Middle: Used for ciphers with multiple stages or keys.
- Ensuring high resistance against these attacks is essential to the cipher's security.

7. Efficiency

- Goal: The cipher should be efficient in both encryption and decryption processes.
- Description: The design should ensure that the cipher runs efficiently on the available hardware or software platforms, with a reasonable tradeoff between speed and security. It should minimize computational complexity, as excessive rounds or operations can reduce performance.

8. Randomization

- Goal: Ensure that the same plaintext encrypts to different ciphertexts each time.
- Description: Some ciphers use a technique called initialization vectors (IVs) or nonce values to ensure that the encryption of the same plaintext with the same key results in different cipher texts. This ensures that patterns in the plaintext don't repeat in the cipher text and enhances security.



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9. Minimal Key Size

- Goal: Use the smallest possible key size while maintaining security.
- Description: The key size must be large enough to provide security against brute-force attacks, but it should not be unnecessarily large as that would lead to inefficiency in terms of memory and computation.

10. Block Size

- Goal: Choose a reasonable block size for the cipher to balance security and efficiency.
- Description: A larger block size generally provides better security, as it reduces the likelihood of collision in the cipher text. However, it also increases the processing time for encryption and decryption. Most modern block ciphers use a block size of 128 bits, which provides a reasonable balance between security and performance.

11. Avalanche Effect

- Goal: A small change in the plaintext or the key should drastically change the cipher text.
- Description: This is related to both confusion and diffusion, where a small change in the input (e.g., a single bit in the plaintext) should result in a large, unpredictable change in the output cipher text. This ensures that attackers cannot easily guess patterns in the cipher text.

Q.5 Apply Playfair cipher technique to encrypt the message "MY NAME IS SKIT" by. Use keyword "PLAYFAIR SECURITY".

Ans. To encrypt the message "MY NAME IS SKIT" using the Playfair cipher technique with the keyword "PLAYFAIR SECURITY," we follow these steps:

Create the Playfair matrix/key table using the keyword.

Break the plaintext into digraphs (pairs of two letters).

Apply the Playfair encryption rules to each digraph.

The encryption process:

Create the Playfair matrix/key table:

P L A Y F

I R S E C

U T D G H

K B M N O

Q V W X Z



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Break the plaintext into digraphs: "MY", "NA", "ME", "IS", "SK", "IT".

Apply the Playfair encryption rules to each digraph:

"MY": M → row 3, column 2, Y → row 4, column 4. Encrypted as "MU YO".

"NA": N → row 4, column 1, A → row 1, column 3. Encrypted as "RU VA".

"ME": M → row 3, column 2, E → row 2, column 3. Encrypted as "MO VE".

"IS": I → row 1, column 1, S → row 2, column 2. Encrypted as "PI SE".

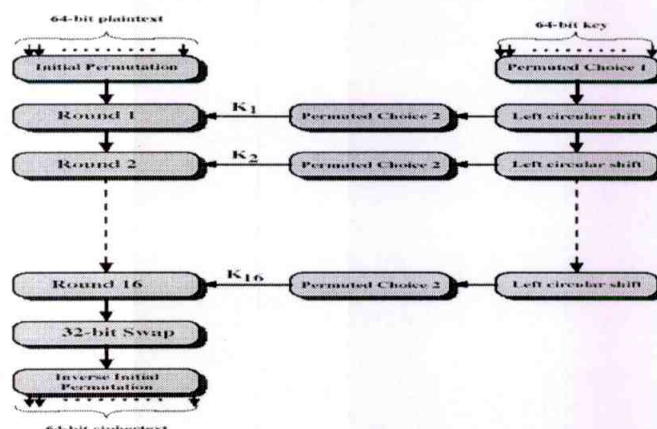
"SK": S → row 2, column 2, K → row 4, column 1. Encrypted as "IS RU".

"IT": I → row 1, column 1, T → row 3, column 3. Encrypted as "PI DG".

So, the encrypted message "MY NAME IS SKIT" using the Playfair cipher technique with the keyword "PLAYFAIR SECURITY" is "MU YO RU VA MO VE PI SE IS RU PI DG".

Q.6 Describe DES algorithm with the help of suitable block diagrams. Compute the 4-bit output of 6-bit input (101101) using S-Box.

Ans. DES (data Encryption standard) is block cipher symmetric key algo. In which 32-bit swap swaps left and 32 bit halves obtained after Round 16, we get pre output. Finally, pre output passes through a permutation IP^{-1} , that is inverse to initial permutation IP, to produce the 64-bit cipher text. The right-hand portion of Fig. 3.1 shows the way in which 56-bit is used. For each of 16 rounds a subkey K_i is produced by the combination of a left circular shift and a permutation. The permutation function is the same for each round.



General Depiction of DES Encryption Algorithm



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Initial Permutation: It affects on 64-bit input

2	34	26	18	10	2
4	36	28	20	12	4
6	38	30	22	14	6
8	40	32	24	16	8
1	33	25	17	9	1
3	35	27	19	11	3
5	37	29	21	13	5
7	39	31	23	15	7

Details of Single Round

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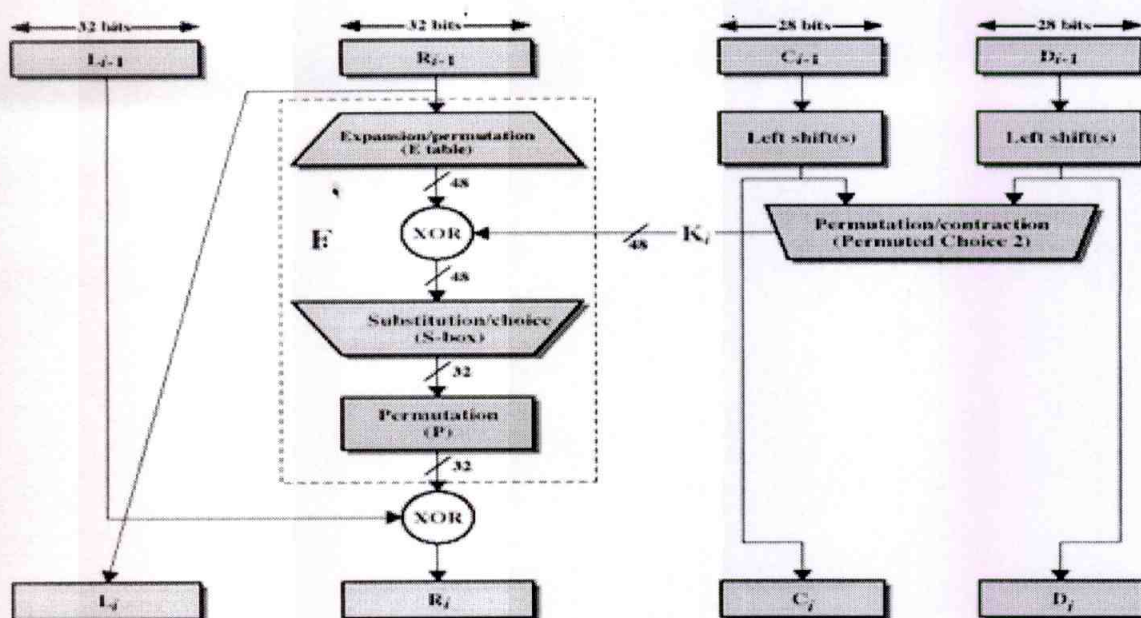


Figure Single Round of DES Algorithm

The left and right halves of each 64-bit intermediate value are treated as separate 32-bit quantities, labeled L and R. As in the classic Feistel cipher, the overall process at each round is summarized as follows:

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

The round key K_i is 48 bits. The R input is 32 bits. This R input is first expanded to 48 bits by Expansion/Permutation (E table):

Expansion/Permutation (E table)					
32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25



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24	25 26 27 28	29
28	29 30 31 32	1

The resulting 48 bits are XORed with K_i . This 48 bit result passes through a substitution function that produces 32-bit output which is permuted by Permutation function (P):

The role of S-boxes is illustrated in Fig.

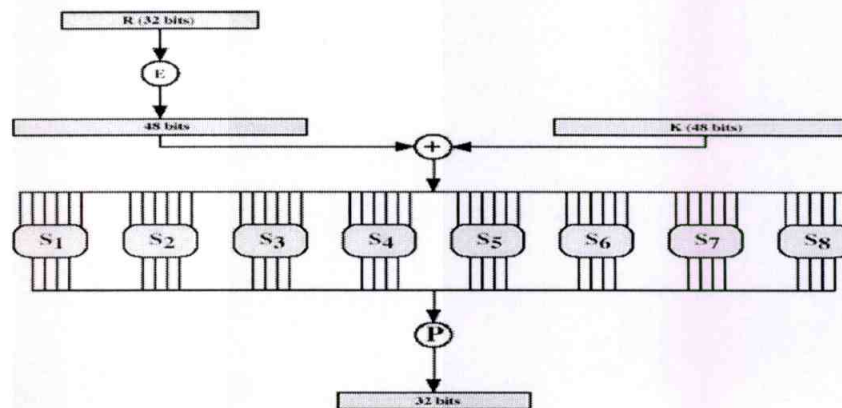


Figure Calculation of $F(R, K)$

The substitution consists of a set of 8 S-boxes, each of which accepts 6 bits input and produces 4 bits as output. Each row of an S-box defines a general reversible substitution: middle 4 bits of each group of 6-bit input are substituted by S-box output 1st and last 6th bits define what particular substitution out of to use.

Key Generation

Input key has 64 bits. But each 8th bit is not used: bits 8,16,24,32,40,48,56,64 are not further used. The 56-bit key is first subjected to permutation Permuted Choice 1:

Permuted Choice 1 (PC-1)
57 49 41 33 25 17 9
1 58 50 42 34 26 18
10 2 59 51 43 35 27



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19 11 3 60 52 44 36
63 55 47 39 31 23 15
7 62 54 46 38 30 22
14 6 61 53 45 37 29
21 13 5 28 20 12 4

The resulting 56-bit key is then treated as 2 28-bit quantities, labeled C0 and D0. At each round, C_{i+1} and D_{i+1} are separately subjected to a circular left shift, or rotation, of 1 or 2 bits as governed by the following:

Schedule of Left Shifts																
Round number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bits rotated	1	1	2	2	2	2	2	1	2	2	2	2	2	2	2	1

These shifted values serve as input to the next round. They also serve as input to Permuted Choice 2, which produces a 48-bit output that serves as input to the function $F(R_{i-1}, K_i)$.

Permuted Choice 2 (PC-2)																
14	17	11	24	1	5	3	28									
15	6	21	10	23	19	12	4									
26	8	16	7	27	20	13	2									
41	52	31	37	47	55	30	40									
51	45	33	48	44	49	39	56									
34	53	46	42	50	36	29	32									

Des Decryption

As with any Feistel cipher, decryption uses the same algorithm as encryption, except that the application of sub-keys is reversed.

I/P: 101101 $\rightarrow$ (b1,b6) $\rightarrow$ 1 1 $\rightarrow$ 3 (in binary) ROW number
(b2 to b5) $\rightarrow$ 0110 $\rightarrow$ 6 (in binary) COLUMN number

Intersection $\rightarrow$ 4 $\rightarrow$ 0100 (in binary)



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Solution of Question Paper
I Mid-Term Examination, March -2025

Prog./Sem.: B. Tech.-VI	Course Title: ISS	Course Code: 6IT4-03
Duration: 1.5 hours	Date: 11 March, 2025 Session : Second	Max Marks: 20
Submitted By: Ritu Shukla , Sunita Gupta		

Q.7. Discuss the concept of key distribution in symmetric key cryptography. Compute the shared secret key using the Diffie-Hellman Key Exchange Algorithm with the provided values:

Given: Prime number $p=5$, Base $g=3$, Alice's private key $a=6$, Bob's private key $b=15$.

Ans. In symmetric key cryptography, both parties must use the same secret key for encryption and decryption. The challenge is securely distributing this key between them.

The methods of key distribution are as follows:

1. **Pre-shared Key (PSK):** The key is physically shared in advance or via a secure channel.
2. **Key Distribution Center (KDC):** A trusted third party provides keys dynamically to communicating entities (e.g., Kerberos).
3. **Public Key Cryptography:** Symmetric keys can be exchanged securely using asymmetric encryption (e.g., RSA or Diffie-Hellman).
4. **Diffie-Hellman Key Exchange:** A mathematical method to securely generate a shared secret over an insecure channel.
 - Prime number $p = 5$
 - Base $g = 3$
 - Alice's private key $a = 6$
 - Bob's private key $b = 15$

Step 1: Compute Public Keys

Each party computes their **public key** using the formula:

$$A = g^a \mod p$$

$$B = g^b \mod p$$

For Alice: $A = 3^6 \mod 5$

$$A = 729 \mod 5 = 4$$

For Bob: $B = 3^{15} \mod 5$

Step 2: Compute the Shared Secret

Now, both Alice and Bob compute the shared secret using the received public key:

For Alice:

$$S = B^a \mod p = 36 \mod 5 = 1$$

For Bob:



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$$S = A^b \text{ mod } p = 4^{15} \text{ mod } 5$$

Thus, the **shared secret key** is 4.



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I-Mid Term Examination, March-2025

Prog./Semester:	B. Tech.-V	Branch:	IT
Course Title:	CAO	Course Code:	6IT4-04
Duration:	1.5 Hours	Maximum Marks:	20
Session:	Second	Roll No.:	
Instructions if any:			

PART -A (short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	Define (r-1)'s and r's complement using example	2	1	L1
2	Define register transfer language and microoperation.	2	1	L1
3	Define rules of writing a assembly language program.	2	2	L1

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	Draw and explain flow chart of first pass of two pass assembler.	4	2	L2
5	A digital computer has a common bus system for 4 registers of 4 bit each. The bus is constructed with MUX. 1. How many selection inputs are there in each mux. 2. What size of mux are needed. 3. How many mux are needed.	4	1	L2

PART- C (Descriptive /Analytical/Problem solving/Design questions)
(Attempt any one Question)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
6	What are the different types of instruction formats based on number of addresses? Explain. (or) Evaluate the arithmetic statement $X = (A+B) * (C+D)$ using a general register computer with three address, and two address, one address and zero address instruction format?	6	3	L3
7	Discuss all factors which effects the performance of pipeline processor based system. A non pipeline system takes 120 ns to complete a task. The same task can be processed in a six segment pipeline with a clock cycle of 20 ns. Determine the speedup ratio of the pipeline system for 100 tasks. What is maximum speed that can be achieved?	6	3	L3



Solution of Question Paper
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Prog./Sem.: B. Tech.-VI	Course Title: CAO	Course Code: 6IT4-04
Duration: 1.5 hours	Date: 10.3.2025	Max Marks: 20
Submitted By: Manju Choudhary		

S. No.	Course Outcome	Percentage Coverage
CO1	Understand the basic operations of complex digital computer systems, arithmetic logic and shift micro operations in symbolic form at a register transfer level	40%
CO2	Apply the assembly language concepts, microcode generation for the control memory and addressing modes used in instructions	30%
CO3	Analyze clock periods, performance, and instruction throughput of single-cycle, multi-cycle, and pipelined implementations of a simple instruction set.	30%
CO4	Apply algorithms for arithmetic operations and implementation for ALU design.	-
CO5	Classify different memory organizations.	-

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	30%
2	BL2- Understand	40%
3	BL3- Apply	30%
4	BL4- Analyze	-
5	BL5- Evaluate	-
6	BL6- Create	-

-- Question Paper Solution --

Ans 1: Define (r-1)'s and r's complement using example

The mostly used complements are 1's, 2's, 9's, and 10's complement. Apart from these complements, there are many more complements from which mostly peoples are not familiar. For finding the subtraction of the number base system, the complements are used. If r is the base of the number system, then there are two types of complements that are possible, i.e., r's and (r-1)'s. We can find the r's complement, and (r-1)'s complement of the number, here r is the radix. The r's complement is also known as Radix complement (r-1)'s complement, is known as Diminished Radix complement.

If the base of the number is 2, then we can find 1's and 2's complement of the number. Similarly, if the number is the octal number, then we can find 7's and 8's complement of the number.

There is the following formula for finding the r's and (r-1)'s complement:

$$r' \text{ s complement} = (r^n)_{10} - N$$

$$(r-1)' \text{ s complement} = \{(r^n)_{10} - 1\} - N$$

In the above formulas,

- The n is the number of digits in the number.



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Solution of Question Paper
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Duration: 1.5 hours	Date: 10.3.2025 Session : Second	Max Marks: 20
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- The N is the given number. The r is the radix or base of the number

Ans 2: Define register transfer language and microoperation.:

Register Transfer Language (RTL) is a low-level language that is used to describe the functioning of a digital circuit and, more specifically, the transfer of information between registers. It provides how data moves from one register to the other and how data is processed within the digital system. Through RTL, there is a capability of creating abstraction levels where high-level design descriptions can be created and easily linked to low-level hardware implementation in designing, simulating, as well as synthesizing digital circuits.

The symbolic notation used to describe the micro-operation transfer among registers is called RTL (Register Transfer Language). → The use of symbols instead of a narrative explanation provides an organized and concise manner for listing the micro-operation sequences in registers and the control functions that initiate them.

Micro-operations

The operation executed on the data store in registers are called micro-operations. They are detailed low-level instructions used in some designs to implement complex machine instructions.

Types of Micro-operations:

- Register Transfer Micro-operations: Transfer binary information from one register to another. →
- Arithmetic Micro-operations: Perform arithmetic operation on numeric data stored in registers.
- Logical Micro-operations: Perform bit manipulation operations on data stored in registers.
- Shift Micro-operations: Perform shift operations on data stored in registers.

Ans 3: Define rules of writing a assembly language program:

- A programming language is defined by a set of rules.
- The basic unit of an assembly language program is a line of code.
- set of rules specify the symbols that can be used and how they may be combined to form a line of code.
- Each line of an assembly language program is arranged in three columns called fields.
- The fields specify the following information.



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1. The field may be empty or it may specify a symbolic address.
2. The label instruction field specifies a machine instruction or a pseudo instruction.
3. The comment field may be empty or it may include a comment

A symbolic address consists of one, two, or three, but not more than three alphanumeric characters.

The instruction field in an assembly language program may specify one of the following items:

1. A memory-reference instruction (MRI)
2. A register-reference or input-output instruction (non-MRI)
3. A pseudo instruction with or without an operand
 - A memory-reference instruction occupies two or three symbols separated by spaces. The first must be a three-letter symbol defining an MRI operation. The second is a symbolic address. The third symbol, which may or may not be present, is the letter.
 - non-MRI is defined as an instruction that does not have an address part.
 - A pseudoinstruction is not a machine instruction but rather an instruction to the assembler giving information about some phase of the translation.

Ans 4: Draw and explain flow chart of first pass of two pass assembler:

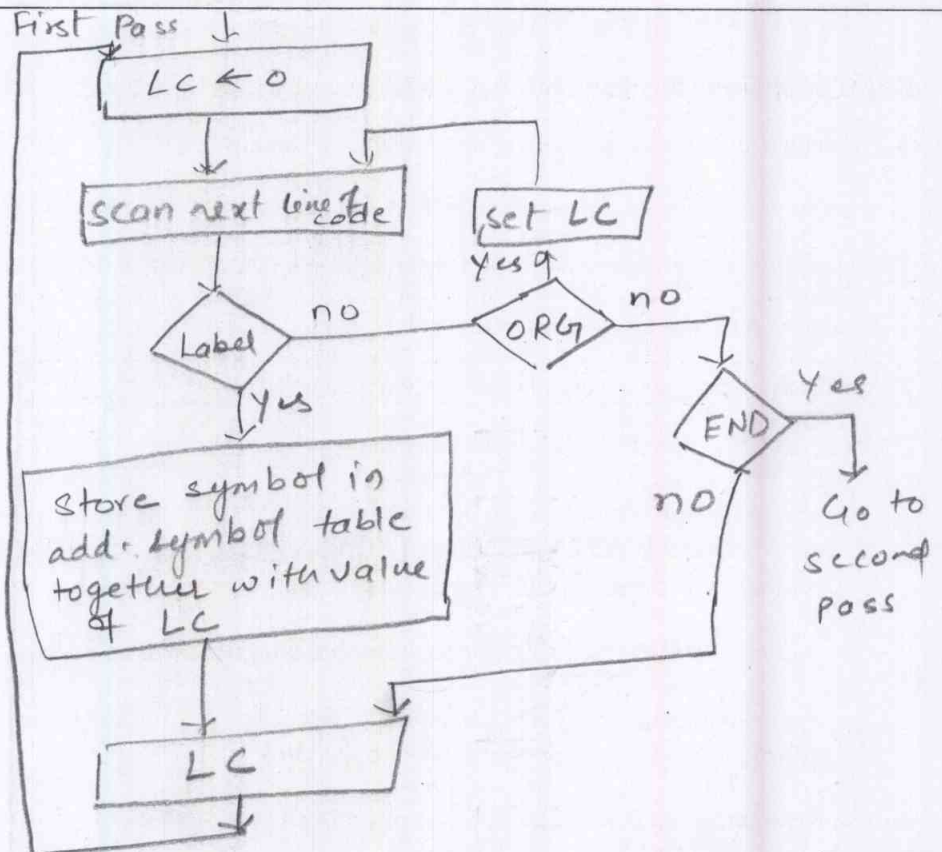
A two-pass assembler scans the entire symbolic program twice. During the first pass, it generates a table that correlates all user-defined address symbols with their binary equivalent value. The binary translation is done during the second pass. To keep track of the location of instructions, the assembler uses a memory location counter (LC) word called a location counter (abbreviated LC). The content of LC stores the value of the memory location assigned to the instruction or operand presently being processed. The ORG pseudoinstruction initializes the location counter to the value of the first location. Since instructions are stored in sequential

locations, the content of LC is incremented by 1 after processing each line of code. To avoid ambiguity in case ORG is missing, the assembler sets the location counter to 0 initially. LC is initially set to 0. A line of symbolic code is analyzed to determine if it has a label (by the presence of a comma). If the line of code has no label, the assembler checks the symbol in the instruction field. If it contains an ORG pseudoinstruction, the assembler sets LC to the number that follows ORG and goes back to process the next line. If the line has an END pseudoinstruction, the assembler terminates the first pass and goes to the second pass. (Note that a line with ORG or END should not have a label.) If the line of code contains a label, it is stored in the address symbol table together with its binary equivalent number specified by the content of LC. Nothing is stored in the table if no label is encountered. LC is then incremented by 1 and a new line of code is processed.



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Flow chart for first Pass of assembles

Ans 5: A digital computer has a common bus system for 4 registers of 4 bit each. The bus is constructed with MUX.

1. How many mux are needed
2. What size of mux are needed
3. How many selection inputs are there in each mux

1. MUX are needed: no of bits in each register

$$\text{No of Mux} = 4$$

2. Size of mux= No of registers*1

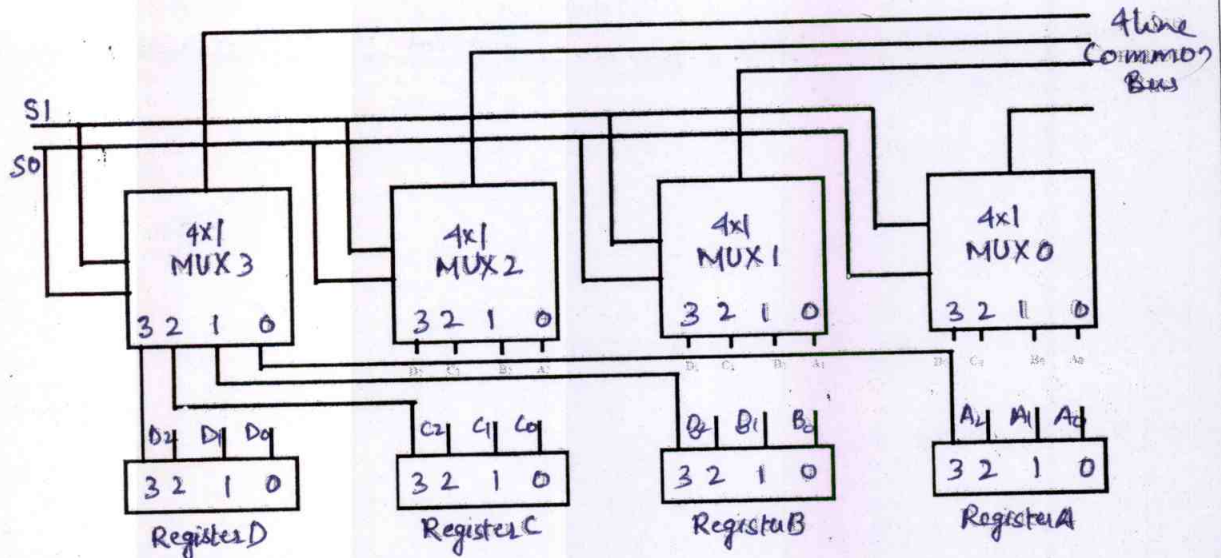
$$= 4 * 1$$

3. Selection inputs: $4 = 2^2$



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Ans 6: What are the different types of instruction formats based on number of addresses? Explain. (or)
Evaluate the arithmetic statement $X = (A+B) * (C+D)$ using a general register computer with three address, and two address, one address and zero address instruction format?

Types of Instructions: Based on the number of addresses, instructions are classified as:

Zero Address Instructions

These instructions do not specify any operands or addresses. Instead, they operate on data stored in registers or memory locations implicitly defined by the instruction. For example, a zero-address instruction might simply add the contents of two registers together without specifying the register names.

One Address Instructions

These instructions specify one operand or address, which typically refers to a memory location or register. The instruction operates on the contents of that operand, and the result may be stored in the same or a different location. For example, a one-address instruction might load the contents of a memory location into a register.

Two Address Instructions

These instructions specify two operands or addresses, which may be memory locations or registers. The instruction operates on the contents of both operands, and the result may be stored in the same or a different location. For example, a two-address instruction might add the contents of two registers together and store the result in one of the registers.

Three Address Instructions

These instructions specify three operands or addresses, which may be memory locations or registers. The instruction operates on the contents of all three operands, and the result may be stored in the same or a different location. For example, a three-address instruction might multiply the contents of two registers together and add the contents of a third register, storing the result in a fourth register.

$$X = (A+B) * (C+D)$$



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PUSH	A	$TOS \leftarrow A$
PUSH	B	$TOS \leftarrow B$
ADD		$TOS \leftarrow (A + B)$
PUSH	C	$TOS \leftarrow C$
PUSH	D	$TOS \leftarrow D$
ADD		$TOS \leftarrow (C + D)$
MUL		$TOS \leftarrow (C + D) * (A + B)$
POP	X	$M[X] \leftarrow TOS$

One address Instruction:

LOAD	A	$AC \leftarrow M[A]$
ADD	B	$AC \leftarrow A[C] + M[B]$
STORE	T	$M[T] \leftarrow AC$
LOAD	C	$AC \leftarrow M[C]$
ADD	D	$AC \leftarrow AC + M[D]$
MUL	T	$AC \leftarrow AC * M[T]$
STORE	X	$M[X] \leftarrow AC$

Two address instruction:

MOV	R1, A	$R1 \leftarrow M[A]$
ADD	R1, B	$R1 \leftarrow R1 + M[B]$
MOV	R2, C	$R2 \leftarrow M[C]$
ADD	R2, D	$R2 \leftarrow R2 + M[D]$
MUL	R1, R2	$R1 \leftarrow R1 * R2$
MOV	X, R1	$M[X] \leftarrow R1$

Three address instruction:

ADD	R1, A, B	$R1 \leftarrow M[A] + M[B]$
ADD	R2, C, D	$R2 \leftarrow M[C] + M[D]$
MUL	X, R1, R2	$M[X] \leftarrow R1 * R2$

Ans 7: Factors which effects the performance of pipeline processor:

Factors are as follows:

- 1) Pipeline latency: The fact that the execution time of each instruction does not decrease puts limitations on pipeline depth.
- 2) Imbalance among pipeline stages: Imbalance among the pipe stages reduce performance since the clock can



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run no faster than the time needed for the slowest pipeline stages.

- 3) Pipeline overhead: Pipeline overhead arises from the combination of pipeline register delay and clock skew.

Time non pipelined system $t_n = 100\text{ns}$

Time taken pipelined system $t_p = 20\text{ ns}$

No of segment $k = 6$

No of task $n = 200$

we get here first time taken by Non-Pipeline to process 100 task is

Time taken for non Pipeline Processor = $200 \times 100 = 20000\text{ ns}$

so here Time Taken by Pipe lined Processor will be

Time Taken Pipe line Processor = $(k + n - 1) \times t_p$

put here value and we will get

Time Taken Pipe line Processor = $(6 + 200 - 1) \times 20$

****Time Taken Pipe line Processor = 4100 ns**

so here **speedup** will be = Non pipeline/ Pipeline

$$= 20000/4100 = 4.87$$

Max speed up that can be achieved = $100/20$
= 5



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I-Mid Term Examination, March-2025

Prog./Semester:	B. Tech.-VI	Branch:	IT
Course Title:	AI	Course Code:	6IT4-05
Duration:	1.5 Hours	Maximum Marks:	20
Session : First		Roll No.:	
Instructions if any:			

PART - A (short answer type questions)

PART -A (short answer type questions)

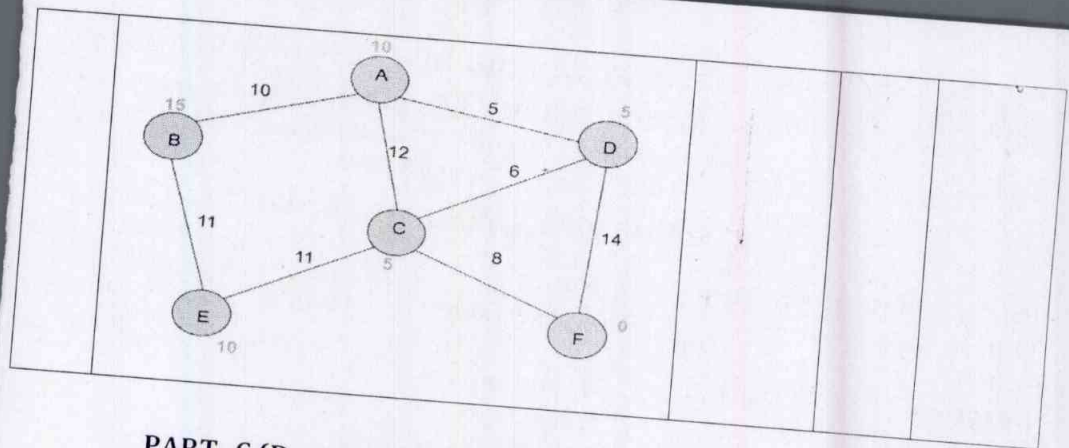
(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	Explain Artificial Intelligence? List out some of task domains of Artificial Intelligence	2	CO1	BL2
2	Explain Constraint Satisfaction with suitable example.	2	CO1	BL2
3	Discuss Game Playing with its types	2	CO2	BL2

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	Analyze and differentiate between Informed and Uninformed Search algorithms by categorizing various searching techniques based on their characteristics and heuristics.	4	CO2	BL4
5	Solve the following example using A* Algo (Min cost from S to D)	4	CO1	BL3



PART- C (Descriptive/Analytical/Problem solving/Design questions)(Attempt any one Question)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
6	You are given 2 jugs, 4 litre capacities and 3 litre capacities. Neither has any Measuring mark on it. We have unlimited water supply. How can we get exactly 2 Litre water in 4 litre jug? Show using state space search equation.	6	CO2	BL3
7	Calculate the optimal sequence of moves for following example using Alpha Beta pruning. Analyze time complexity of Minimax Algorithm and Alpha Beta pruning.	6	CO2	BL3



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**Solution of Question Paper
I Mid-Term Examination, March-2025**

Prog./Sem.: B. Tech.-VI	Course Title: AI	Course Code:6IT4-05
Duration: 1.5 hours	Date: 10/03/2025	Max Marks: 20
Submitted By: Ms Shalini Singhal		

S. No.	Course Outcome	Percentage Coverage
CO1	Understand the informed and uninformed problem types and apply search strategies to solve them.	60%
CO2	Apply intelligent algorithms for constraint satisfaction problems and also design intelligent systems for Game Playing.	40%
CO3	Apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and reasoning.	-
CO4	Illustrate basic principle of AI in expert System and Natural Language Processing.	-
CO5	Analyze the limitations of different type of machine Learning model.	-

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	-
2	BL2- Understand	30%
3	BL3- Apply	50%
4	BL4- Analyze	20%
5	BL5- Evaluate	-
6	BL6- Create	-

-- Question Paper Solution -

1. Give Outline to the Artificial Intelligence? List out some of task domains of Artificial Intelligence?

The ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings. The term is frequently applied to the project of developing systems endowed with the intellectual processes characteristic of humans, such as the ability to reason, discover meaning, generalize, or learn from past experience.

Task Domain of AI:--

1. AI in Astronomy

Artificial Intelligence can be very useful to solve complex universe problems. AI technology can be helpful for understanding the universe such as how it works, origin, etc.

2. AI in Healthcare

Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

3. AI in Gaming

AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.

4. AI in Finance

AI and finance industries are the best matches for each other. The finance industry is implementing automation, chatbot, adaptive intelligence, algorithm trading, and machine learning into financial processes.

5. AI in Data Security



Solution of Question Paper
I Mid-Term Examination, March-2025

Prog./Sem.: B. Tech.-VI	Course Title: AI	Course Code:6IT4-05
Duration: 1.5 hours	Date: 10/03/2025	Max Marks: 20
Submitted By: Ms Shalini Singhal		

The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure. Some examples such as AEG bot, AI2 Platform, are used to determine software bug and cyber-attacks in a better way.

6. AI in Social Media

Social Media sites such as Facebook, Twitter, and Snapchat contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyze lots of data to identify the latest trends, hashtag, and requirement of different users.

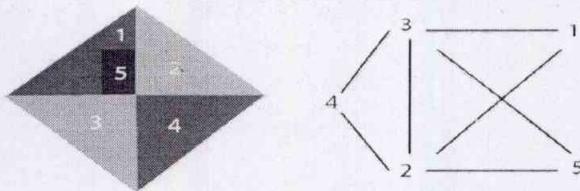
2. Explain Constraint Satisfaction with suitable example.?

Constraint satisfaction is a technique where a problem is solved when its values satisfy certain constraints or rules of the problem. Such type of technique leads to a deeper understanding of the problem structure as well as its complexity.

Constraint satisfaction depends on three components, namely:

- X: It is a set of variables.
- D: It is a set of domains where the variables reside. There is a specific domain for each variable.
- C: It is a set of constraints which are followed by the set of variables.

Graph Coloring: The problem where the constraint is that no adjacent sides can have the same color.



Graph Coloring

3. Discuss Game Playing with its types?

Searches in which two or more players with conflicting goals are trying to explore the same search space for the solution, are called adversarial searches, often known as Games.

Games are modeled as a Search problem and heuristic evaluation function, and these are the two main factors which help to model and solve games in AI.

Types of Games in AI:

o Perfect information: A game with the perfect information is that in which agents can look into the complete board. Agents have all the information about the game, and they can see each other moves also. Examples are Chess, Checkers, Go, etc.

o Imperfect information: If in a game agents do not have all information about the game and not such as tic-tac-toe, Battleship, blind, Bridge, etc.

o Deterministic games: Deterministic games are those games which follow a strict pattern and set of rules for the games, and there is no randomness associated with them. Examples are chess, Checkers, Go, tic-tac-toe, etc.

o Non-deterministic games: Non-deterministic are those games which have various unpredictable events and has a factor of chance or luck. This factor of chance or luck is introduced by either dice or cards. These are random, and each action response is not fixed. Such games are also called as stochastic games. Example: Backgammon, Monopoly, Poker, etc. Demonstrate Spiral Model with its applications.



Solution of Question Paper
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Prog./Sem.: B. Tech.-VI	Course Title: AI	Course Code: 6IT4-05
Duration: 1.5 hours	Date: 10/03/2025 Session : FIRST	Max Marks: 20
Submitted By: Ms Shalini Singhal		

4. Analyze and differentiate between Informed and Uninformed Search algorithms by categorizing various searching techniques based on their characteristics and heuristics?

Uninformed search is a class of general-purpose search algorithms which operates in brute force-way. Uninformed search algorithms do not have additional information about state or search space other than how to traverse the tree, so it is also called blind search.

Following are the various types of uninformed search algorithms:

- Breadth-first Search
- Depth-first Search
- Depth-limited Search
- Iterative deepening depth-first search
- Uniform cost search
- Bidirectional Search

Informed search is a type of search algorithm in artificial intelligence that uses additional information or heuristics to make more accurate decisions about which paths to explore first. These heuristics provide estimates of how close a given state is to the goal, guiding the search toward more promising solutions. Informed search is particularly useful in solving complex problems efficiently, as it can significantly reduce the search space and improve the speed of finding solutions.

Informed search algorithms can quickly reject irrelevant or less promising alternatives, allowing the search to concentrate on the most reliable options, by employing domain-specific knowledge to drive the search. Heuristics are used by these kinds of AI search algorithms to increase the search's effectiveness and speed.

Best First Search

The path that appears to be the best at the time is always chosen by the greedy best-first search algorithm. It combines techniques like depth-first search and breadth-first search. It uses search and the heuristic function. We can benefit from both algorithms' advantages by using best-first search. We can select the most promising node at each stage using a best-first search. The node that is nearest to the goal node is expanded using the best first search technique, and the closest cost is calculated using a heuristic function.

We expand the node that is closest to the goal node, and the closest cost is estimated by a Heuristic function, i.e. $f(n) = h(n)$.

Where, $h(n)$ = estimated cost from node n to the goal.

A* Search

A* search is the most commonly known form of best-first search. The heuristic function $h(n)$, along with the distance from the initial state $g(n)$ to the node n , is used. Due to the combination of UCS and greedy best-first search features, the issue is effectively solved. Using the heuristic function, the A* search method locates the shortest route through the search space.

Hill Climbing

To ascertain the summit of a mountain or to identify the optimal solution to a given problem, one may employ the hill climbing algorithm. This algorithm, categorized as a local search technique, perpetually advances in the direction of elevated terrain or increased value. When it reaches a peak value where none of its neighbors have a greater value, it ends.

The hill climbing algorithm is a method for solving mathematical problems with optimization. Traveling-salesman is one of the most frequently mentioned cases of a hill climbing algorithm problem where we need to reduce the salesman's journey distance.

5. Solve the following example using A* Algo (Min cost from S to D)?



Solution of Question Paper
I Mid-Term Examination, March-2025

Prog./Sem.: B. Tech.-VI	Course Title: AI	Course Code: 6IT4-05
Duration: 1.5 hours	Date: 10/03/2025	Max Marks: 20
Submitted By: Ms Shalini Singhal		

- Graph Representation – The given graph has weighted edges.
- Heuristic Function ($h(n)$) – The estimated cost from a node to the goal.
- Cost Function ($g(n)$) – The actual cost from the start node to the current node.
- Evaluation Function ($f(n)$) – Defined as: $f(n) = g(n) + h(n)$
 - $g(n)$: Cost to reach node n from the start.
 - $h(n)$: Estimated cost from n to the goal.

Step-by-Step Solution

1. Define Given Elements

- Nodes: A, B, C, D, E, F
- Edges with Weights: Provided in the image.
- Start Node: B
- Goal Node: F
- $h(n)$ values: Given as the orange numbers in the image.

2. Initialize

- We maintain an Open List (nodes to explore) and a Closed List (nodes already explored).

Node Heuristic $h(n)$

A	10
B	15
C	5
D	5
E	10
F	0

3. Start from Node B

- $g(B) = 0$ (Cost to start node itself)
- $f(B) = g(B) + h(B) = 0 + 15 = 15$
- Add B to Open List, set Parent(B) = None.

4. Expand Node B (Neighbors: A, C, E)

- A: $g(A) = g(B) + \text{cost}(B \rightarrow A) = 0 + 10 = 10$
 $f(A) = g(A) + h(A) = 10 + 10 = 20$
- C: $g(C) = g(B) + \text{cost}(B \rightarrow C) = 0 + 11 = 11$
 $f(C) = g(C) + h(C) = 11 + 5 = 16$
- E: $g(E) = g(B) + \text{cost}(B \rightarrow E) = 0 + 11 = 11$
 $f(E) = g(E) + h(E) = 11 + 10 = 21$

lowest $f(n)$: C (16).

5. Expand Node C (Neighbors: A, D, E, F)

- A: Already exists, but $g(A) = 11 + 12 = 23$ (worse than previous 20), so ignore.
- D: $g(D) = g(C) + \text{cost}(C \rightarrow D) = 11 + 6 = 17$
 $f(D) = g(D) + h(D) = 17 + 5 = 22$
- E: Already exists, new $g(E) = 11 + 11 = 22$ (worse than 21), so ignore.
- F: $g(F) = g(C) + \text{cost}(C \rightarrow F) = 11 + 8 = 19$
 $f(F) = g(F) + h(F) = 19 + 0 = 19$

lowest $f(n)$: F (19).



Solution of Question Paper
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Submitted By: Ms Shalini Singhal		

6. Goal Reached!

The path found is $B \rightarrow C \rightarrow F$ with a total cost of 19.

Final Path Output

Optimal Path: $B \rightarrow C \rightarrow F$

Total Cost: 19

6. You are given 2 jugs, 4 litre capacities and 3 litre capacities. Neither has any measuring mark on it. We have unlimited water supply. How can we get exactly 2 litre water in 4 litre jug? Show using state space search equation.

The Water Jug Problem is a classic state-space search problem in Artificial Intelligence. We have:

- Jug 1 (4L capacity)
- Jug 2 (3L capacity)
- Goal: Obtain exactly 2L in the 4L jug
- Unlimited water supply and emptying is allowed

State Representation

Each state is represented as (x, y) where:

- x = amount of water in the 4L jug.
- y = amount of water in the 3L jug.

State Space Search Approach

We use the Breadth-First Search (BFS) or Depth-First Search (DFS) to explore all possible operations:

Allowed Operations

1. Fill Jug 1 (4L jug) $\rightarrow (4, y)$
2. Fill Jug 2 (3L jug) $\rightarrow (x, 3)$
3. Empty Jug 1 $\rightarrow (0, y)$
4. Empty Jug 2 $\rightarrow (x, 0)$
5. Pour Jug 1 $\rightarrow$ Jug 2 (until Jug 2 is full or Jug 1 is empty)
6. Pour Jug 2 $\rightarrow$ Jug 1 (until Jug 1 is full or Jug 2 is empty)

Step-by-Step Solution

Using BFS, we explore the valid states:

Initial State: $(0,0)$

1. Fill 3L jug $\rightarrow (0,3)$
2. Pour 3L into 4L jug $\rightarrow (3,0)$
3. Fill 3L jug again $\rightarrow (3,3)$
4. Pour from 3L jug into 4L jug until full $\rightarrow (4,2)$
5. Empty 4L jug $\rightarrow (0,2)$ (Goal Reached!)

State Space Search Equations

The state transition function $f(s, a) \rightarrow s'$ follows:

1. Initial State: $S_0 = (0,0)$ $S_{-0} = (0,0)$ $S_0 = (0,0)$
2. Goal State: $S_g = (2,y)$ $S_{-g} = (2,y)$ $S_g = (2,y)$
3. Operators:
 - $f((x,y), \text{Fill_4}) = (4,y)$ $f((x,y), \text{Fill_4}) = (4,y)$
 - $f((x,y), \text{Fill_3}) = (x,3)$ $f((x,y), \text{Fill_3}) = (x,3)$



Solution of Question Paper
I Mid-Term Examination, March-2025

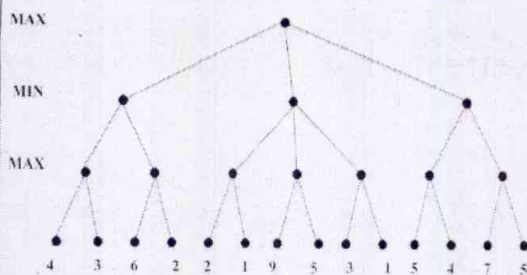
Prog./Sem.: B. Tech.-VI	Course Title: AI	Course Code:6IT4-05
Duration: 1.5 hours	Date: 10/03/2025	Max Marks: 20
Submitted By: Ms Shalini Singhal		

- $f((x,y), \text{Empty_4}) = (0,y) f((x,y), \text{Empty_4}) = (0,y) f((x,y), \text{Empty_4}) = (0,y)$
- $f((x,y), \text{Empty_3}) = (x,0) f((x,y), \text{Empty_3}) = (x,0) f((x,y), \text{Empty_3}) = (x,0)$
- $f((x,y), \text{Pour_4_to_3}) f((x,y), \text{Pour_4_to_3}) f((x,y), \text{Pour_4_to_3})$
- $f((x,y), \text{Pour_3_to_4}) f((x,y), \text{Pour_3_to_4}) f((x,y), \text{Pour_3_to_4})$

Solution Path

$(0,0) \rightarrow (0,3) \rightarrow (3,0) \rightarrow (3,3) \rightarrow (4,2) \rightarrow (0,2) (0,0) \rightarrow (0,3) \rightarrow (3,0) \rightarrow (3,3) \rightarrow (4,2) \rightarrow (0,2)$
 $\rightarrow (0,2) (0,0) \rightarrow (0,3) \rightarrow (3,0) \rightarrow (3,3) \rightarrow (4,2) \rightarrow (0,2)$

7. Calculate the optimal sequence of moves for following example using Alpha Beta pruning. Analyze time complexity of Minimax Algorithm and Alpha Beta pruning.



Alpha (best value for MAX) and Beta (best value for MIN) help cut off unnecessary computations.

Traversal and Pruning Process

- Step 1: Evaluate the first MAX subtree $(4,3) \rightarrow$ MAX chooses 4.
- Step 2: Evaluate the next subtree $(6,2) \rightarrow$ MAX chooses 6.
- Step 3: MIN chooses the minimum between 4 and 6 $\rightarrow$ 4.
- Step 4: Evaluate the next MAX subtree $(2,1) \rightarrow$ MAX chooses 2.
- Step 5: Evaluate the next subtree $(9,5) \rightarrow$ MAX chooses 9.
- Step 6: MIN chooses $\min(2,9) = 2$.
- Step 7: Since 4 is already smaller than 2, we prune the rest of the tree on this branch.
- Step 8: Repeat for the remaining branches and prune where unnecessary.

Final Minimax Value

- The optimal move sequence is determined at the root by traversing the pruned tree.
- The final Minimax value at the root is the best move for MAX.

Time Complexity Analysis

Minimax Algorithm Complexity



Solution of Question Paper
I Mid-Term Examination, March-2025

Prog./Sem.: B. Tech.-VI	Course Title: AI	Course Code: 6IT4-05
Duration: 1.5 hours	Date: 10/03/2025	Max Marks: 20
Submitted By: Ms Shalini Singhal		
Session : FIRST		

- If the tree has b branches per node and d depth,
- Time Complexity: $O(bd)O(b^d)O(bd)$
- This is exponential, making it inefficient for large-depth trees.

Alpha-Beta Pruning Complexity

- Best case (when pruning is maximally effective): $O(bd/2)O(b^{d/2})O(bd/2)$
- Reduces complexity significantly compared to standard Minimax.

Conclusion

- Without pruning: $O(bd)O(b^d)O(bd)$
- With pruning: $O(bd/2)O(b^{d/2})O(bd/2)$ (approximately square root of original complexity)
- Pruning helps skip unnecessary evaluations, improving efficiency.

-----Finish-----



**Swami Keshvanand Institute of Technology,
Management & Gramothan, Jaipur
I -Mid Term Examination, March-2025**

Prog./Semester:	B. Tech.-VI	Branch:	IT
Course Title:	DS	Course Code:	6IT4-06
Duration:	1.5 Hours	Maximum Marks:	20
Session: First	Roll No.:		
Instructions if any:	N/A		

PART -A (short answer type questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	List the main characteristics of a DS?	2	1	1
2	Describe Distributed Computing Environment? Name any four components of DCE.	2	1	2
3	Define Process and Threads. Write two similarities and two differences between these two.	2	2	1

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
4	Compare and contrast the Network Operating System and Distributed Operating System.	4	1	4
5	Analyse the graph models for Process Representation in Distributed System.	4	2	4

PART- C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any one Question)

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
6	Discuss the language mechanism for synchronization in distributed systems.	6	2	4
7	Describe architecture of RPC and RMI in DS. Write a case study of any one (either RPC or RMI).	6	2	4



Solution of Question Paper
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Prog./Sem.: B.Tech. VI sem	Course Title: Distributed Systems	Course Code: 6IT4-06
Duration: 1.5 hours	Date: 1/03/2025	Max Marks: 20
Session: FIRST		
Submitted By: Dr. Saroj Agrawal/ Ms. Sanju Choudhary		

S. No.	Course Outcome	Percentage Coverage
C01	Understand the fundamental concepts associated with distributed systems. And various issues.	40%
C02	Illustrate and analyze process management, Inter-process Communication techniques.	60%
C03	Compare Distributed File Systems, and Distributed Process Scheduling with Transaction Services and Concurrency Control.	NA
C04	Analyze Distributed Shared Memory with memory consistency models and models of distributed computation.	NA
C05	Discover different algorithms and techniques for fault tolerant, concepts of distributed agreement and replicated data management.	NA

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	20%
2	BL2- Understand	10%
3	BL3- Apply	NA
4	BL4- Analyze	70%
5	BL5- Evaluate	NA
6	BL6- Create	NA



Solution of Question Paper
I Mid-Term Examination, March -2025

Prog./Sem.: B.Tech. VI sem	Course Title: Distributed Systems	Course Code: 6IT4-06
Duration: 1.5 hours	Date: 1/03/2025 Session: FIRST	Max Marks: 20
Submitted By: Dr. Saroj Agrawal/ Ms. Sanju Choudhary		

-- Question Paper Solution--

Q. No.	Question	Max. Marks	CO(s)	Bloom's Level
1	List the main characteristics of a DS?	2	1	1
2	Describe Distributed Computing Environment? Name any four components of DCE.	2	1	2
3	Define Process and Threads. Write two similarities and two differences between these two.	2	2	1
4	Compare and contrast the Network Operating System and Distributed Operating System.	4	1	4
5	Discuss the graph models for Process Representation in Distributed Systems.	4	2	4
6	Discuss the language mechanism for synchronization in distributed systems.	6	2	4
7	Describe architecture of RPC and RMI in DS. Write a case study of any one (either RPC or RMI).	6	2	4

A1> (any 4 can be considered) Concurrency: Distributed systems involve multiple independent components running concurrently, allowing for parallel processing and multitasking.

Transparency: Distributed systems aim to provide transparency to users, hiding the complexities of the underlying architecture. This includes transparency in access, location, migration, replication, concurrency, and failure.

Scalability: Distributed systems are designed to scale horizontally, allowing for the addition of more resources or nodes to handle increasing workloads without significant performance degradation.

Fault Tolerance: Distributed systems incorporate mechanisms to tolerate failures, ensuring that the system continues to function despite individual component failures. This includes redundancy, replication, and fault detection and recovery mechanisms.

Interoperability: Distributed systems support interoperability, allowing components developed using different technologies and programming languages to communicate and work together seamlessly.

Heterogeneity: Distributed systems often comprise heterogeneous hardware and software components, including different types of operating systems, programming languages, and network protocols.

Autonomy: Distributed systems consist of autonomous components that operate independently and make local decisions without central coordination. This autonomy enhances system flexibility and responsiveness.

Resource Sharing: Distributed systems facilitate resource sharing among multiple users or applications, enabling



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efficient utilization of computing resources such as processing power, storage, and bandwidth.

Consistency: Distributed systems aim to maintain data consistency across multiple nodes, ensuring that all replicas of data remain synchronized and up-to-date.

Security: Distributed systems implement security measures to protect data and resources from unauthorized access, ensuring confidentiality, integrity, and availability.

A2> Distributed Computing Environment (DCE). The Distributed Computing Environment (DCE) is a set of specifications and technologies developed by the Open Software Foundation (OSF) in the late 1980s and early 1990s. DCE was designed to provide a standardized framework for building and deploying distributed computing systems across heterogeneous hardware and software platforms.

components of the Distributed Computing Environment include-(any 4 can be considered)

Remote Procedure Call (RPC):

Security Services:

Directory Services:

Naming Services:

Time Services:

Threads and Concurrency Control:

Interoperability and Portability:

A3> Process- A process is defined as a sequence of instructions executed in a predefined order. In simple words, any program that is executed is termed as a process. Processes change their state as they execute and can be either new, ready, running, waiting or terminated.

Thread- A thread is a flow of execution through the process code, with its own program counter that keeps track of which instruction to execute next, system registers which hold its current working variables, and a stack which contains the execution history.

Any 2 can be considered for both

Similarities: •Like processes, threads share CPU and only one thread is active (running) at a time. •Like processes, threads within a process, execute sequentially. •Like processes, thread can create children. •And like a process, if one thread is blocked, another thread can run.

Differences: •Unlike processes, threads are not independent of one another. •Unlike processes, all threads can access every address in the task. •Unlike processes, threads are designed to assist one another. Note that processes might or might not assist one another because processes may originate from different users.

A4> Compare and contrast the Network Operating System and Distributed Operating System.



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Submitted By: Dr. Saroj Agrawal/ Ms. Sanju Choudhary		

3. Comparison: Network OS vs. Distributed OS

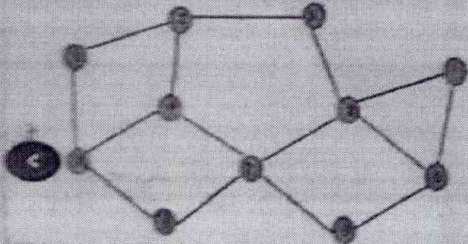
Feature	Network OS (NOS)	Distributed OS (DOS)
Definition	Manages resources in a networked environment	Manages resources across multiple systems as a single unit
Autonomy	Each node runs independently	Nodes work in collaboration
Resource Sharing	Users explicitly share resources	Resources are shared transparently
Fault Tolerance	Limited fault tolerance	High fault tolerance with redundancy
Example	Windows Server, Linux	Amoeba, Mach, LOCUS

Bloom Level: Evaluating (Level 5)

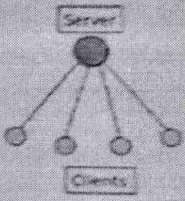
Ans 5 Discuss the graph models for Process Representation in Distributed System.

- Graph representation
 - Nodes = processes
 - Edges = communication links *channels*
 - Links can be bidirectional (undirected graph) or unidirectional (directed graph)
 - Links can have weights to represent different things (Ex. delay, length, bandwidth,...)
 - Links in the graph may or may not correspond with physical links

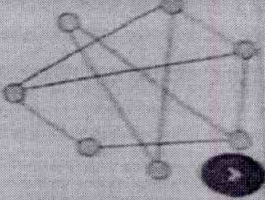
Topology of a DS is represented by a graph



Examples



Client-server model
Server is the coordinator



Peer-to-peer model
No unique coordinator



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Session: FIRST		
Submitted By: Dr. Saroj Agrawal/ Ms. Sanju Choudhary		

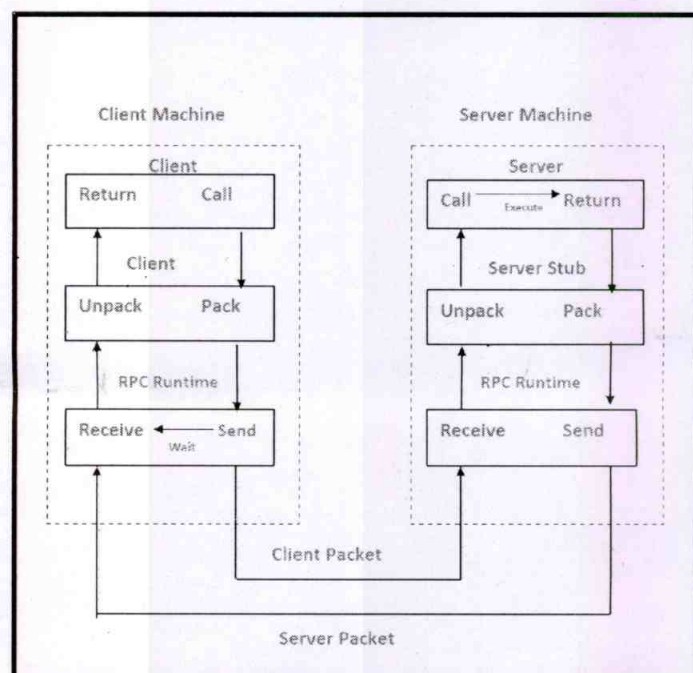
A6> Discuss the language mechanism for synchronization in distributed systems.

Synchronization mechanisms ensure proper sequencing of events in distributed systems. Some language-based mechanisms include:

1. **Monitors:** Encapsulate shared data and provide synchronization primitives.
2. **Message Passing:** Uses send() and receive() primitives for inter-process communication.
3. **Semaphores & Mutexes:** Control access to shared resources using wait (P()) and signal (V()) operations.
4. **Condition Variables:** Used in monitors to delay processes until a specific condition is met.

A 7. Describe architecture of RPC and RMI in DS. Write a case study of any one (either RPC or RMI).

Remote Procedure Call (RPC) – It is a communication paradigm that has an ability to invoke a new process or a method on a remote process as if it were a local procedure call. The client process makes a procedure call using RPC and then the message is passed to the required server process using communication protocols. These message passing protocols are abstracted and the result once obtained from the server process, is sent back to the client process to continue execution.



The RMI system consists of three layers: the *stub/skeleton layer*, the *remote reference layer*, and the *transport layer*. The boundary at each layer is defined by a specific interface and protocol; each layer, therefore, is independent of the next



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Submitted By: Dr. Saroj Agrawal/ Ms. Sanju Choudhary		

and can be replaced by an alternate implementation without affecting the other layers in the system. For example, the current transport implementation is TCP-based (using Java sockets), but a transport based on UDP could be substituted.

To accomplish transparent transmission of objects from one address space to another, the technique of object serialization (designed specifically for the Java language) is used. Object serialization is described in this chapter only with regard to its use for marshaling primitives and objects. For complete details, see the specification *Object Serialization in the Java System*.

Another technique, called *dynamic stub loading*, is used to support client-side stubs which implement the same set of remote interfaces as a remote object itself. This technique, used when a stub of the exact type is not already available to the client, allows a client to use the Java language's built-in operators for casting and type-checking.

Architectural Overview

The RMI system consists of three layers:

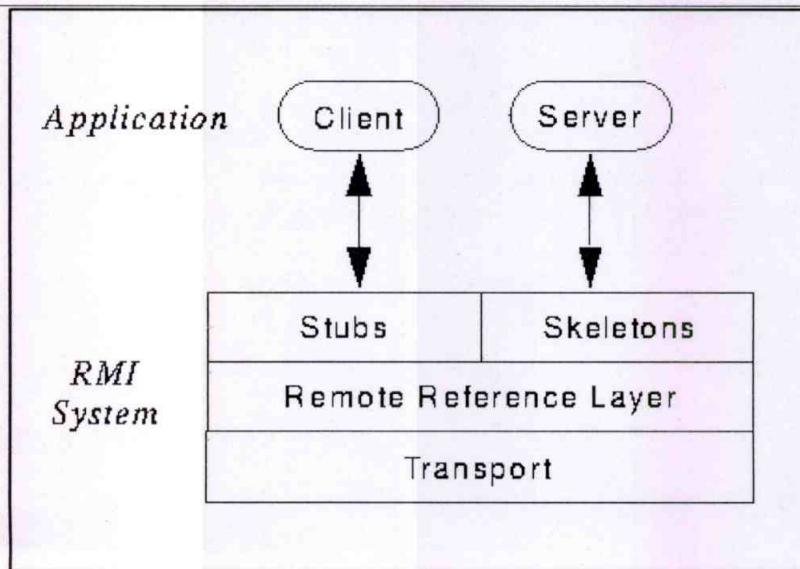
- The stub/skeleton layer - client-side stubs (proxies) and server-side skeletons
- The remote reference layer - remote reference behavior (e.g. invocation to a single object or to a replicated object)
- The transport layer - connection set up and management and remote object tracking

The application layer sits on top of the RMI system. The relationship between the layers is shown in the following figure.



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Submitted By: Dr. Saroj Agrawal/ Ms. Sanju Choudhary		



A remote method invocation from a client to a remote server object travels down through the layers of the RMI system to the client-side transport, then up through the server-side transport to the server.

A client invoking a method on a remote server object actually makes use of a *stub* or proxy for the remote object as a conduit to the remote object. A client-held reference to a remote object is a reference to a local stub. This stub is an implementation of the remote interfaces of the remote object and forwards invocation requests to that server object via the remote reference layer. Stubs are generated using the `rmic` compiler.

The *remote reference layer* is responsible for carrying out the semantics of the invocation. For example, the remote reference layer is responsible for determining whether the server is a single object or is a replicated object requiring communications with multiple locations. Each remote object implementation chooses its own remote reference semantics—whether the server is a single object or is a replicated object requiring communications with its replicas.

Also handled by the remote reference layer are the reference semantics for the server. The remote reference layer, for example, abstracts the different ways of referring to objects that are implemented in (a) servers that are always running on some machine, and (b) servers that are run only when some method invocation is made on them (activation). At the layers above the remote reference layer, these differences are not seen.

The *transport* is responsible for connection set-up, connection management, and keeping track of and dispatching to remote objects (the targets of remote calls) residing in the transport's address space.

In order to dispatch to a remote object, the transport forwards the remote call up to the remote reference layer. The



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Submitted By: Dr. Saroj Agrawal/ Ms. Sanju Choudhary		

remote reference layer handles any server-side behavior that needs to be done before handing off the request to the server-side skeleton. The skeleton for a remote object makes an up-call to the remote object implementation which carries out the actual method call.

The return value of a call is sent back through the skeleton, remote reference layer and transport on the server side, and then up through the transport, remote reference layer and stub on the client side.

Java RMI example:

- A banking system where a client remotely accesses account details.
- The client sends a request to the RMI registry, which locates the appropriate remote object on the server.
- The server executes the method and returns the result to the client.



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Duration: 1.5 hours	Date: 1/03/2025 Session: FIRST	Max Marks: 20
Submitted By: Dr. Saroj Agrawal/ Ms. Sanju Choudhary		



Swami Keshvanand Institute of Technology, Management & Gramothan, Jaipur

I -Mid Term Examination, March-2025

Prog./Semester:	B. Tech.-VI	Branch:	IT
Course Title:	Cloud Computing	Course Code:	6IT5-12
Duration:	1.5 Hours	Maximum Marks:	20
Session : Second		Roll No.:	
Instructions if any:		NA	

PART -A (short answer type questions)
(All questions are compulsory)

Q.	Question	Max. Marks	CO(s)	Bloom's Level
1	Brief the term cloud computing along with its objectives.	2	1	1
2	What are the advantages/main features of cloud infrastructure over conventional physical infrastructure?	2	1	1
3	Explain different challenges faced in Cloud Computing technology.	2	2	2

PART -B (Analytical/Problem solving questions)
(All questions are compulsory)

Q.	Question	Max. Marks	CO (s)	Bloom's Level
4	Determine the benefits of migrating Financial Institution Y's legacy banking application to the cloud. Discuss the potential risks associated with the migration and recommend best practices to ensure a smooth transition.	4	1	3
5	Examine how adhering to the principles outlined in the cloud reference model can benefit Start-up Y In designing and deploying their SaaS platform. Identify key components of the reference model that are relevant to Start-up Y's requirements and explain how they contribute to scalability, security, and performance.	4	2	4

PART- C (Descriptive/Analytical/Problem solving/Design questions)
(Attempt any one Question)

Q.	Question	Max. Marks	CO (s)	Bloom's Level
6	Analyse how MapReduce in the cloud can help E-commerce Platform Y analyse website traffic patterns and user behaviour to improve performance and scalability. Discuss the architecture and implementation considerations for deploying real-time analytics using MapReduce on cloud platforms.	6	2	4
7	Outline the potential benefits of each cloud service model (IaaS, PaaS, SaaS) for Financial Institution Y's migration project. Discuss how each service model aligns with the institution's goals for modernization, security, and compliance with regulatory requirements.	6	2	4



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Solution of Question Paper
I Mid-Term Examination, March -2025

Prog./Sem.: B.tech/VI	Course Title: Cloud Computing	Course Code: 6IT5-12
Duration: 1.5 hours	Date: 08 March,2025 Session : SECOND	Max Marks: 20
Submitted By: Dr. Anil Chaudhary, Mr. Jagendra Singh Chaudhary		

S. No.	Course Outcome	Percentage Coverage
C01	Understand the fundamental principles and architecture of cloud computing	40%
C02	Analyze various cloud service models (IaaS, PaaS, SaaS).	60%
C03	Identify security, privacy, and compliance issues in cloud environments.	-
C04	Discover scalable and resilient cloud-based applications.	-
C05	Evaluate economic and business implications of cloud computing solutions	-

S. No.	Bloom's Level	Percentage Coverage
1	BL1- Remember	20%
2	BL2- Understand	10%
3	BL3- Apply	20%
4	BL4- Analyze	50%
5	BL5- Evaluate	-
6	BL6- Create	-

-- Question Paper Solution --

PART -A (short answer type questions)
(All questions are compulsory)

Q1: Brief the term cloud computing along with its objectives.

ANS: Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale. You typically pay only for cloud services you use, helping you lower your operating costs, run your infrastructure more efficiently, and scale as your business needs change.

Cloud computing aims to provide on-demand, scalable, and efficient computing resources. **The key objectives include:**

- Cost Efficiency
- Scalability & Flexibility
- Accessibility & Collaboration
- Reliability & Business Continuity
- Security & Compliance
- Automation & Self-Service
- Energy Efficiency & Sustainability



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Solution of Question Paper
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Q2: What are the advantages/main features of cloud infrastructure over conventional physical infrastructure?

ANS: Cloud infrastructure offers several advantages over traditional on-premises physical infrastructure, making it a preferred choice for businesses and individuals.

1. Cost Efficiency

Physical infrastructure requires high initial investment and ongoing maintenance costs.

2. Scalability & Flexibility

Physical infrastructure is limited by available hardware and requires manual upgrades, which are costly and time-consuming.

3. High Availability & Reliability

Physical infrastructure is prone to failures (e.g., hardware crashes, power outages), leading to data loss and downtime.

4. Accessibility & Remote Work Support

Physical infrastructure requires on-site presence, limiting accessibility and collaboration.

5. Security & Compliance

Physical infrastructure requires dedicated security teams and custom security implementations, increasing complexity and cost.

6. Faster Deployment & Automation

Physical infrastructure deployment takes weeks or months due to hardware procurement and setup.

7. Energy Efficiency & Sustainability

Physical infrastructure requires dedicated power and cooling, increasing operational costs.

8. Global Reach & Content Delivery

Physical infrastructure is limited to a single location, leading to high latency for distant users.

Q3: Explain different challenges faced in Cloud Computing technology.

ANS: Cloud computing offers numerous benefits, but it also presents several challenges that organizations must address. Some of the key challenges include:

1. Security and Privacy Concerns: Security remains a top concern for businesses moving to the cloud. Issues such as data breaches, unauthorized access, and compliance with data protection regulations can arise. Ensuring data privacy and implementing robust security measures are critical.

2. Data Protection and Compliance: Different regions have varying data protection regulations, such as GDPR in Europe or HIPAA in the United States. Organizations must ensure compliance with these regulations when storing and processing data in the cloud.

3. Data Sovereignty: Data stored in the cloud may be subject to the laws and regulations of the country where the data is physically located. This can raise concerns about data sovereignty, especially when using cloud services hosted in multiple regions.

4. Downtime and Reliability: Despite high uptime guarantees provided by cloud service providers, downtime can still occur due to various factors such as network outages, hardware failures, or software bugs. Ensuring reliable access to cloud services is essential for business continuity.

5. Vendor Lock-In: Moving to a specific cloud provider can lead to vendor lock-in, making it difficult to migrate services or data to another provider. Interoperability standards and careful planning can help mitigate this risk.



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6. Performance and Latency: Performance issues and latency can occur when accessing cloud services, especially if the data centers are located far from the users. Optimizing network configurations and utilizing content delivery networks (CDNs) can help improve performance.

7. Cost Management: While cloud computing offers scalability and cost-effectiveness, it's essential to monitor and manage cloud usage to avoid unexpected expenses. Cloud costs can quickly escalate if resources are not provisioned and managed efficiently.

8. Complexity of Integration: Integrating cloud services with existing on-premises systems and applications can be complex. Compatibility issues, data migration challenges, and the need for specialized skills may arise during the integration process.

9. Data Transfer Bottlenecks: Transferring large volumes of data to and from the cloud can be time consuming and costly, especially over the internet. Employing efficient data transfer mechanisms and considering the use of dedicated network connections can help mitigate this challenge.

10. Resource Allocation and Scalability: Properly allocating resources and scaling cloud infrastructure to meet changing demand patterns can be challenging. Adopting automated scaling mechanisms and closely monitoring resource utilization can help optimize performance and costs.

PART -B (Analytical/Problem solving questions)

(All questions are compulsory)

Q4: Determine the benefits of migrating Financial Institution Y's legacy banking application to the cloud. Discuss the potential risks associated with the migration and recommend best practices to ensure a smooth transition.

ANS:

1) Benefits of Cloud Migration for Financial Institution Y

Migrating a legacy banking application to the cloud provides several advantages:

a) Cost Efficiency

Lower Infrastructure Costs – No need to maintain expensive on-premises servers.

Pay-as-you-go Pricing – Financial Institution Y pays only for the resources used.

b) Scalability & Performance

Auto-Scaling – The cloud can handle increased transaction loads during peak times.

Improved Performance – Cloud data centers use high-performance computing and storage.

c) Security & Compliance

Advanced Security – Cloud providers offer encryption, firewalls, and identity management.

Regulatory Compliance – Cloud services comply with financial regulations (ISO 27001, PCI-DSS, GDPR).

d) Business Continuity & Disaster Recovery

High Availability – Cloud ensures 99.9% uptime using redundancy and failover mechanisms.

Backup & Disaster Recovery – Automated backups protect against data loss.

e) Accessibility & Innovation

Remote Banking Services – Employees and customers can securely access banking applications from anywhere.

AI & Machine Learning Integration – Enables fraud detection, risk management, and data analytics.



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2) Potential Risks of Cloud Migration

Despite the benefits, migrating financial applications to the cloud comes with risks:

a) Data Security & Privacy Risks

Risk: Sensitive financial data could be exposed during migration.

Mitigation: Use end-to-end encryption and secure access controls (IAM, MFA).

b) Downtime & Service Disruptions

Risk: Migration errors can cause banking services to become temporarily unavailable.

Mitigation: Perform staged migration and use load balancing to minimize disruptions.

c) Compliance & Regulatory Challenges

Risk: Financial data must comply with banking regulations.

Mitigation: Choose cloud providers that meet PCI-DSS, GDPR, ISO 27001, and local banking laws.

d) Vendor Lock-In

Risk: Dependency on a single cloud provider can limit flexibility.

Mitigation: Use multi-cloud or hybrid cloud strategies to avoid lock-in.

3) Best Practices for a Smooth Cloud Migration

Step 1: Cloud Readiness Assessment

Evaluate the legacy system architecture, dependencies, and compliance requirements.

Determine whether a public, private, or hybrid cloud is suitable.

Step 2: Choose the Right Migration Strategy

Migration Strategy	Description	When to Use?
Rehosting (Lift & Shift)	Move application without modification	Quick migration with minimal changes
Replatforming	Optimize for cloud performance	If small modifications can improve performance
Refactoring	Redesign application for cloud-native architecture	Best for modernizing legacy systems

Step 3: Data Security & Compliance

Encrypt data in transit and at rest.

Implement IAM roles, Multi-Factor Authentication (MFA), and access controls.

Conduct regular security audits.

Step 4: Conduct a Phased Migration

Start with non-critical workloads before migrating core banking services.

Use parallel environments (on-premises + cloud) to minimize downtime.

Step 5: Continuous Monitoring & Optimization

Implement real-time monitoring for performance and security.

Use Auto-scaling and Load Balancing to handle traffic spikes.



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Q5: Examine how adhering to the principles outlined in the cloud reference model can benefit Start-up Y In designing and deploying their SaaS platform. Identify key components of the reference model that are relevant to Start-up Y's requirements and explain how they contribute to scalability, security, and performance.

ANS: A Cloud Reference Model provides a structured approach for designing and deploying cloud-based solutions, ensuring that Start-up Y's Software-as-a-Service (SaaS) platform is scalable, secure, and high-performing.

1. Key Benefits of Adhering to the Cloud Reference Model for Start-up Y

1.1. Scalability

- Elastic Compute & Storage: Start-up Y can scale its computing and storage needs dynamically based on demand.
- Load Balancing & Auto-Scaling: Ensures smooth performance even under high traffic.
- Micro services Architecture: Allows independent scaling of different components.

1.2. Security

- Identity & Access Management (IAM): Ensures secure access control to the SaaS platform.
- Data Encryption: Protects sensitive customer data at rest and in transit.
- Compliance & Governance: Aligns with regulatory requirements (GDPR, ISO 27001, etc.).

1.3. Performance & Reliability

- Multi-Region Deployment: Reduces latency by placing servers closer to users.
- Content Delivery Network (CDN): Speeds up content distribution for global users.
- Continuous Monitoring & Logging: Detects and mitigates performance bottlenecks.

2. Key Components of the Cloud Reference Model & Their Role in Start-up Y's SaaS Platform

Component Relevance to Start-up Y Contribution to Scalability, Security & Performance

Infrastructure as a Service (IaaS)	Provides computing, storage, and networking on demand.	Enables auto-scaling, flexible resource provisioning, and cost efficiency.
Platform as a Service (PaaS)	Offers managed databases, AI services, and DevOps tools.	Simplifies development, enhances security, and accelerates deployment.
Software as a Service (SaaS)	The SaaS platform itself, delivering services to customers.	Ensures availability, global access, and seamless updates.
Security Layer	IAM, Firewalls, Encryption	Ensures data protection, access control, and compliance.
Networking & Connectivity	Load Balancers, CDN, VPN	Enhances latency reduction, availability, and performance.



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3. Implementation Recommendations for Start-up Y

To fully leverage the Cloud Reference Model, Start-up Y should:

- Choose the Right Cloud Provider (AWS, Azure, and GCP) for IaaS & PaaS services.
- Use Managed Databases & Server less computing to optimize costs & performance.
- Implement Zero Trust Security using IAM, MFA, and continuous security monitoring.
- Adopt CI/CD Pipelines for fast deployment & software updates.
- Leverage AI-based Monitoring to detect threats & optimize resources automatically.

Conclusion:

By aligning with the Cloud Reference Model, Start-up Y can build a scalable, secure, and high-performing SaaS platform. This ensures:

- Elastic scaling to handle user growth.
- Robust security to protect sensitive customer data.
- Reliable performance for a seamless user experience.

PART- C (Descriptive/Analytical/Problem solving/Design questions)

(Attempt any one Question)

Q6: Analyze how MapReduce in the cloud can help E-commerce Platform Y analyses website traffic patterns and user behavior to improve performance and scalability. Discuss the architecture and implementation considerations for deploying real-time analytics using MapReduce on cloud platforms.

ANS: E-commerce Platform Y can use MapReduce in the cloud to analyze website traffic patterns and user behavior efficiently. By processing large-scale data distributed across multiple servers, MapReduce enables scalable, high-performance analytics that help improve user experience, optimize marketing strategies, and enhance infrastructure scalability.

1. How MapReduce Benefits E-commerce Platform Y

1.1. Scalability & Performance Optimization

- **Handles Big Data Efficiently:** MapReduce can process massive datasets (e.g., millions of website visits) in parallel, ensuring quick insights.
- **Distributes Workload:** By dividing tasks across multiple cloud-based nodes, it prevents server bottlenecks and speeds up data processing.
- **Auto-Scaling on Cloud:** The cloud automatically provisions resources based on demand, ensuring cost-effective scalability.

1.2. Improved User Behavior Insights

- **Clickstream Analysis:** Tracks which pages users visit, their time spent, and their navigation paths.
- **Session Analysis:** Identifies frequent drop-off points and enhances UI/UX accordingly.
- **Purchase Behavior Prediction:** Detects popular products, abandoned carts, and customer buying patterns to improve recommendations.



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1.3. Real-Time Analytics & Decision Making

- Detect Traffic Spikes in Real Time: Prevents downtime by scaling infrastructure dynamically.
- Fraud Detection & Anomaly Identification: Monitors unusual behavior (e.g., repeated failed login attempts) to prevent cyber threats.
- Personalized Marketing: Segments users based on browsing habits and recommends tailored promotions.

2. MapReduce Architecture for Cloud-Based E-commerce Analytics

MapReduce consists of three main phases:

2.1. Map Phase:

- Input Data (Website logs, user clicks, transaction records) is split into chunks.
- Mappers process each chunk to extract relevant insights (e.g., number of clicks per product).
- Intermediate Key-Value Pairs are generated (e.g., ("Product A", 5000 clicks)).

2.2. Shuffle & Sort Phase:

- Intermediate key-value pairs are sorted and grouped by key before being sent to the reducers.

2.3. Reduce Phase:

- Reducers aggregate the mapped data (e.g., total visits per product).
- The final processed data is stored in cloud-based storage (HDFS, S3, BigQuery) for reporting.

3. Implementation Considerations for Deploying MapReduce on Cloud Platforms

3.1. Choosing the Right Cloud Platform

- AWS: Amazon EMR (Elastic MapReduce)
- Google Cloud: Dataproc
- Azure: HDInsight

3.2. Storage Considerations

- Use distributed storage like HDFS, Amazon S3, or Google Cloud Storage to handle large-scale data efficiently.

3.3. Optimizing MapReduce Performance

- Data Partitioning & Parallel Processing: Ensures faster computation by breaking data into smaller chunks.
- Using Spot Instances for Cost Efficiency: Reduces cost by running jobs on lower-cost instances.
- Compression & Data Locality: Minimizes data transfer overhead.

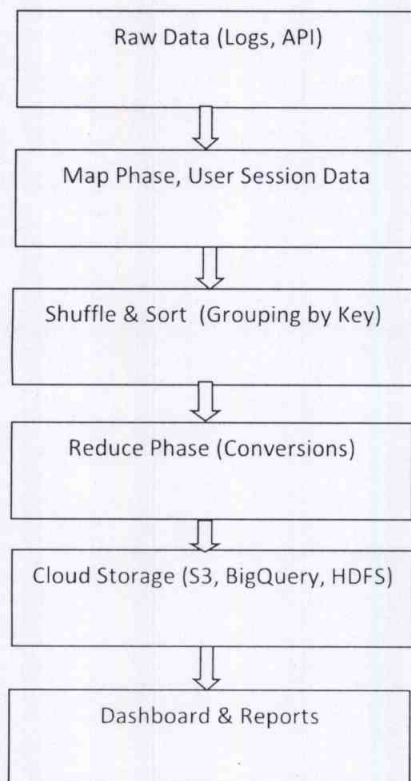


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3.4. Real-Time Analytics with MapReduce Alternatives

- Apache Spark (faster than Hadoop MapReduce) for real-time streaming.
- Kafka + Spark Streaming for live traffic monitoring.



Conclusion

By leveraging MapReduce on cloud platforms, E-commerce Platform Y can:

- Process massive website traffic data efficiently.
- Gain deep insights into user behavior for personalization.
- Scale resources dynamically to handle peak loads.



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Q7: Outline the potential benefits of each cloud service model (IaaS, PaaS, SaaS) for Financial Institution Y's migration project. Discuss how each service model aligns with the institution's goals for modernization, security, and compliance with regulatory requirements.

ANS: As Financial Institution Y moves to the cloud, selecting the right cloud service models will be key to achieving modernization, security, and regulatory compliance. The three primary cloud models—Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS)—offer unique advantages that align with the institution's goals.

1. Infrastructure as a Service (IaaS)

Potential Benefits for Financial Institution Y:

- Scalability & Cost Efficiency – Elastic computing resources ensure cost-effective scaling without the need for on-premises infrastructure.
- Security & Control – Full control over servers, storage, and networking allows custom security implementations.
- Disaster Recovery & High Availability – Redundant data storage and failover mechanisms improve system reliability.

Alignment with Institution's Goals:

- Modernization: Shifts infrastructure to cloud-based data centers (e.g., AWS EC2, Azure VMs).
- Security: Custom firewalls, Identity & Access Management (IAM), and encryption enhance security.
- Regulatory Compliance: Meets PCI-DSS, GDPR, and ISO 27001 standards by allowing financial institutions to implement necessary security controls.

Use Case Example:

Hosting core banking applications and risk analysis systems on virtual machines in the cloud.

2. Platform as a Service (PaaS)

Potential Benefits for Financial Institution Y:

- Faster Development & Deployment – Pre-configured environments enable rapid application development.
- Security & Compliance – Built-in security patches, encryption, and compliance frameworks reduce risk.
- AI & Big Data Integration – Supports advanced analytics, fraud detection, and AI-driven financial insights.



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Alignment with Institution's Goals:

- Modernization: Enables cloud-native application development using frameworks like Azure App Services or Google Cloud App Engine.
- Security: PaaS providers handle security updates, reducing vulnerabilities.
- Regulatory Compliance: Automated logging and auditing capabilities help in compliance reporting.

Use Case Example:

Deploying mobile banking apps, fraud detection models, and AI-driven credit risk assessments.\

3. Software as a Service (SaaS)

Potential Benefits for Financial Institution Y:

- No Infrastructure Management – The cloud provider handles hosting, security, and maintenance.
- Easy Compliance & Security – Leading SaaS providers adhere to strict financial regulations.
- Seamless Updates & Scalability – Automatic software updates ensure compliance with evolving industry standards.

Alignment with Institution's Goals:

- Modernization: Uses SaaS-based CRM, financial reporting, and document management systems (e.g., Salesforce, Microsoft 365).
- Security: Data encryption and built-in access controls protect sensitive customer data.
- Regulatory Compliance: Meets financial industry standards (SOX, PCI-DSS, GDPR, etc.) with minimal configuration effort.

Use Case Example:

Utilizing SaaS-based financial management tools, customer relationship management (CRM), and compliance reporting applications.



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4. Comparison: Choosing the Right Cloud Model

Feature	IaaS (Infrastructure)	PaaS (Platform)	SaaS (Software)
Control	High (Full infrastructure control)	Medium (Limited to app deployment)	Low (Managed by provider)
Customization	Full flexibility	Pre-configured environments	Limited
Security	Custom security policies	Security handled by provider	Built-in security & compliance
Compliance	Requires manual implementation	Built-in compliance tools	Fully compliant with financial regulations
Best for	Hosting core banking & risk analysis systems	Developing AI-driven financial models	Using pre-built financial software (CRM, ERP)

Conclusion

For Financial Institution Y's cloud migration, a hybrid approach combining IaaS, PaaS, and SaaS will provide the best balance between modernization, security, and compliance:

- IaaS for secure, scalable infrastructure to host banking systems.
- PaaS for faster development of fraud detection and AI models.
- SaaS for regulatory compliance tools and customer service platforms.